

Elevated heterozygosity clued atypical reproduction: evidence consistent with hybridogenesis mediating insular occurrence of a narrowly endemic spiny frog

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eLife Assessment

This study presents **valuable** findings regarding a rare mode of reproduction called hybridogenesis in a species pair of frogs. While parts of the study provide **solid** support for the claim of hybridogenesis, other parts are **incomplete** with certain claims being only partially supported, as alternative modes of reproduction cannot be fully ruled out.

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Abstract

Hybridogenesis is a rare reproductive mode found in interspecific hybrids that involves discarding one parental genome during gametogenesis and clonally transmitting the other, with the former renewed by backcrossing. It may in theory mediate species range expansion, since such hybrids can also cross and have pure-species offspring for establishing new populations and are often widespread probably due to hybrid vigor. Being inspired by a magnitude difference between estimates of individual heterozygosity, we collected genome-wide data and evaluated predictions from hybridogenesis of hybrids between the wideranging Asian spiny frog *Quasipaa boulengeri* and a narrow endemic, *Q. robertingeri*. We rejected other atypical modes of reproduction such as parthenogenesis or androgenesis and provided the first evidence consistent with a hypothesis that hybridogenesis mediated the occurrence of individuals far from the species range (~500 km). Despite contributing to early evidence of the only distinct hybridogenetic complex inferred in ten years, individual heterozygosity has likely been an often overlooked variable. The accumulating genome-wide data may serve as a resource for searching for individual heterozygosity clues of atypical reproduction of interspecific hybrid origin. The findings in spiny frogs make a case that genome-wide data provide crucial evidence for updating our knowledge on reproductive-mode evolution of taxa.

1. Introduction

The recent widespread availability of high-throughput sequencing facilitates reliable comparison of individual heterozygosity that requires a large number of loci (Nei, 1978; Mitton & Pierce, 1980). Extraordinarily high individual heterozygosity often reflects a heterogeneous genetic composition (e.g., Green, 1984; Muniz et al., 2020; Zhang et al., 2021). It can serve as a simple genetic clue to possible hybridization and subsequent processes, especially to hybridogenesis that is also referred to as hemiclinal reproduction (Hotz et al., 1999). This mode has been found in interspecific hybrids and involves the elimination of one parental genome during gametogenesis and clonal transmission of the left one, with the former renewed by backcrossing (Dedukh et al., 2017; Lavanchy & Schwander, 2019). As such, a hybridogenetic system is featured by elevated heterozygosities of F₁-hybrid types compared to heterozygosities of the other, pure-species individuals. Partly inspired by heterozygosity patterns, classical, non-social hybridogenesis has been identified in six taxa (Bogart et al., 2007; Lafond et al., 2019; Lavanchy & Schwander, 2019; Suzuki et al., 2020; Majtanova et al., 2021). It occurs in hybrids between the live-bearing fish *Poeciliopsis monacha* and several congeners in North America (Miller & Schultz, 1959; Schultz, 1969; Mateos & Vrijenhoek, 2005). The other three fish taxa include an Iberian minnow complex *Squalius alburnoides* (Carmona et al., 1997; Collares-Pereira & Coelho, 2010; Matos et al., 2019), five *Hypseleotris* gudgeon species in Australia (Bertozzi et al., 2000; Schmidt et al., 2011; Thacker et al., 2022), and three *Hexagrammos* greenling species in coasts of southern Hokkaido and adjacent areas (Kimura-Kawaguchi et al., 2014; Munehara et al., 2016). It has also been found in hybrids between the European marsh frog *Pelophylax ridibundus* and two congeners (Berger, 1967; Turner & Heppich-Turner, 1991; Dufresnes & Mazepa, 2020; Dufresnes et al., 2024) and between stick insects *Bacillus rossius* and *B. grandii* in Sicily (Bullini & Nascetti, 1990; Mantovani et al., 1996; Scavariello et al., 2017; Lavanchy et al., 2024). Although one may share the view that these divergent animals represent a small fraction of hybridogenetic systems (Lavanchy & Schwander, 2019), there have been no new taxa reported for ten years in the era of high-throughput sequencing.

Possibly related to this gap, further evidence than individual heterozygosities is needed to confirm hybridogenesis (Majtánová et al., 2021) or to infer and distinguish it from other systems where hybrids are F₁ types. The latter include hybrid sterility and parthenogenesis, gynogenesis, kleptogenesis, and androgenesis of interspecific hybrid origin. In parthenogenesis, the embryo develops from an unfertilized egg. Gynogenesis also transmits the female genome only but requires sperm from a sympatric sexual species to trigger embryogenesis. In kleptogenesis, the sperm acquired/stolen from a co-occurring sexual species usually serve to trigger embryogenesis only. Occasionally, the sperm's nuclear genome can be incorporated and contribute to the gene pool of an asexual population (Fujita et al., 2020). In androgenesis, the nuclear genome is contributed by the male, while eggs lacking a nucleus are produced by females, or the female genome is eliminated after fertilization (Schwander & Oldroyd, 2016). Genome-wide evidence of parental types and a hybrid type characterized by elevated heterozygosity, ~50/50 ancestry proportions, and a large number of fixed heterozygous loci supports but does not distinguish among these systems. Reflecting the genetic divergence between species, considerable fixed heterozygosity is expected for the F₁-hybrid types of hybridogenesis and other modes lacking interspecific recombination, such as ameiotic parthenogenesis and ameiotic androgenesis (e.g., Vrijenhoek et al., 1978; Mantovani & Scali, 1993; Nascetti et al., 2003; Scali et al., 2003; Schwander & Oldroyd, 2016; Flynn et al., 2017; Unmack et al., 2019; Wachi et al., 2021).

On the other hand, several types of evidence may be useful to distinguish hybridogenesis from the others. First, if an F₁-hybrid type is widespread, hybrid sterility and modes assuming many local hybrid origins can be rejected by a lack of coexistence of hybrids with both parental species.

Second, in lineages exhibiting parthenogenesis, gynogenesis, or kleptogenesis, males are very rare if not absent (Avisé, 2008; Tilquin & Kokko, 2016; Fujita et al., 2020), which is not necessarily the case for hybridogenetic lineages (e.g., Schmidt et al., 2011). Third, when a single or few local hybrid origins are assumed, a prevalent local, nuclear haplotype sharing between F₁-hybrid and parental types is not expected for androgenesis, parthenogenesis, gynogenesis, and kleptogenesis. In these modes by definition a co-distributed parental species has no or only occasional contribution to the nuclear genome (Lampert & Scharl, 2010; Lehtonen et al., 2013; Morgado-Santos et al., 2017). Similarly, assuming a single or few local hybrid origins, a widespread high mitochondrial DNA similarity between co-distributed F₁-hybrid and parental types is not expected for parthenogenesis, gynogenesis, and kleptogenesis (Schlupp, 2005; Bi & Bogart, 2010; Schmidt et al., 2011). A gynogenetic or kleptogenetic lineage is expected to exhibit matrilineal mitochondrial DNA variation/haplotypes different from those of the sperm donor populations (Bogart et al., 2009). In a hybridogenetic system, hybridogens share nuclear and possibly mitochondrial variation with a co-occurring parental species through backcrossing (Doležalková-Kaštánková et al., 2021). Last, the non-hybridogenetic modes, if not combined with for instance a meiotic hybridogenesis-like process that forms homologous haploid gametes (e.g., Arai, 2023; Lafond & Angers, 2024), are not expected to have pure-species offspring. Nevertheless it can be accomplished by hybridogens with eggs and sperm carrying conspecific genomes as found in *Pelophylax* frogs (Vorburger, 2001a, b; Dufresnes & Mazepa, 2020). For clarity, a summary of such different signatures of hybridogenesis and other systems is provided in **Table 1**.

If an identified hybridogenetic system does have pure-species offspring of hybridogens, it may be suitable to address an unexplored possibility: disjunct range expansion of a narrowly endemic species mediated by hybridogenesis. An endemic may be restricted by habitat requirements and/or competition from a widespread relative (e.g., McGlone et al., 2001; Matesanz et al., 2009). By forming a hybridogenetic complex with the latter and taking advantage of the expected heterosis (Bulger & Schultz, 1979; Hotz et al., 1999), the former may establish a remote population from pure-species offspring of hybridogens given ecological opportunity. This can be investigated using a hybridogenetic complex ideally between an endemic with disjunct distribution and a widespread species. Such a pattern of distribution, however, has not been observed in the aforementioned six hybridogenetic systems (e.g., Mantovani et al., 1996; Collares-Pereira & Coelho, 2010; Kimura-Kawaguchi et al., 2014; Dufresnes & Mazepa, 2020; Thacker et al., 2022; IUCN, 2024).

In East Asian spiny frogs, *Quasipaa robertingeri* has a narrow known distribution nested within the range of its widespread close relative *Q. boulengeri* (Che et al., 2009; Fei, 2020; Yan et al., 2021; Frost, 2023). They inhabit subtropical mountain streams and their banks, with *Q. robertingeri* recorded only from a small area at the southern edge of the Sichuan Basin (**Fig. 1**) where sympatric populations of them have been found (Wang et al., 2021). Adults of the two species are considered to be morphologically similar. The tadpole of *Q. robertingeri* can be readily distinguished from those of *Q. boulengeri* and closely related *Q. shini*, *Q. spinosa*, *Q. exilispinosa*, and *Q. verrucospinosa* by the absence of dark transverse markings on dorsal tail (Wu & Zhao, 1995; Fei et al., 2009; Pyron, 2014; Yan et al., 2021). Similar karyotypes of 2n = 26 with five large and eight small chromosome pairs have been reported in *Q. robertingeri* and *Q. boulengeri* (Tan & Wu, 1987; Wu & Zhao, 1995; Qing et al., 2012; Xia et al., 2020). In addition, an interchromosomal reciprocal translocation has been inferred to spread among some *Q. boulengeri* populations along the western edge of the Sichuan Basin but not detected elsewhere (Wang, 2006; Qing et al., 2012).

In a preliminary assessment of the genetic diversity of *Q. boulengeri* with 32 samples, we have observed a bimodal heterozygosity pattern that cannot be explained by reciprocal translocation. Reciprocal translocation may suppress meiotic recombination of the chromosomes involved and eventually result in high levels of heterozygosity (McKim et al., 1988; Haigis & Dove, 2003). Single nucleotide polymorphisms (SNPs) of these samples were obtained from restriction site-

Table 1

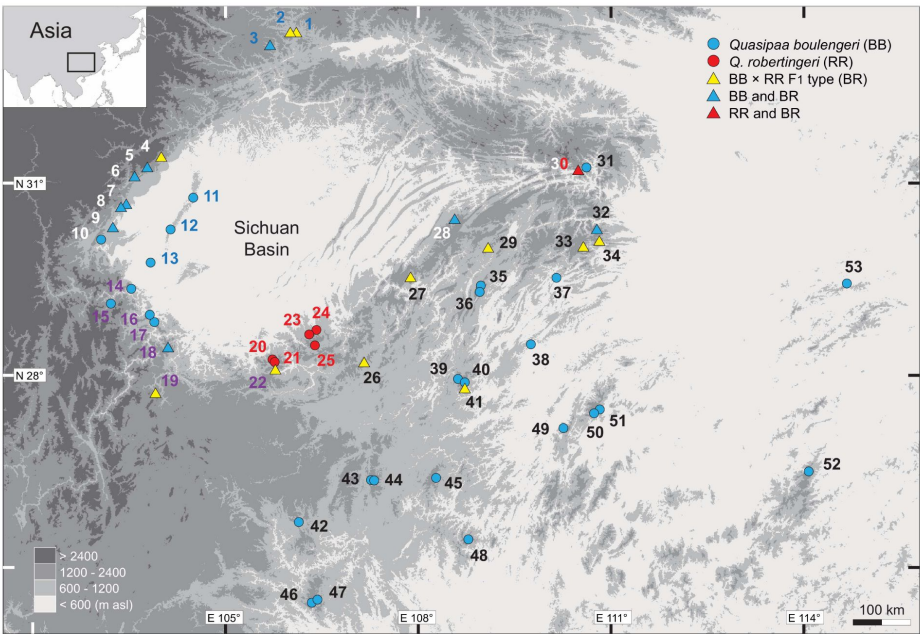
Several types of different signatures of reproductive systems where hybrids are F₁ types.

Feature	Hybrid sterility	Hybridogenesis	Parthenogenesis/gynogenesis/kleptogenesis	Androgenesis
Widespread coexistence of F ₁ type with both parental species	Yes (if F ₁ type widespread)	Yes (if many local hybrid origins)	Yes (if many local hybrid origins)	Yes (if many local hybrid origins)
Male F ₁ type very rare/absent	Not necessarily	Not necessarily	Yes	No
No prevalent local, nuclear haplotype sharing between F ₁ and parental types	No	Unlikely	Yes (if single/a few local hybrid origins)	Yes (if single/a few local hybrid origins)
No widespread mitochondrial variation/haplotype sharing between co-distributed F ₁ and parental types	No	Not necessarily	Yes (for single/a few local hybrid origins)	Not necessarily
No pure-species offspring produced by cross of F ₁ types	Yes	Not necessarily	Yes	Yes

Figure 1

Sampling sites.

Symbols indicate genotype compositions of samples. Site number colors correspond to clusters BB1 and BR1 (black), BB2 and BR2 (purple), BB3 and BR3 (blue), BB4 and BR4 (white), and RR (red) in Fig. 2 [↗](#).



associated DNA sequencing (RAD-seq) data. Some estimates of individual heterozygosity were roughly an order of magnitude higher than the others, and this pattern remained unchanged when analyzing subsets of loci presumably on different chromosomes individually. Moreover, for each of the few loci further examined, a high-heterozygosity sample appeared to have one *Q. boulengeri* haplotype and one outgroup, *Q. robertingeri* haplotype. These results can be explained by that the high-heterozygosity samples are F_1 type hybrids between the two species.

In this study, we obtained RAD-seq data from samples across the distribution of both species and conducted heterozygosity, ancestry proportion, principal coordinate plot, individual clustering, population tree, haplotype network, and mitochondrial gene tree analyses. We first provided evidence supporting a system where hybrids were F_1 types. By integrating our results and the literature, we rejected other atypical reproductive modes and accepted hybridogenesis that could explain the occurrence of pure individuals of *Q. robertingeri* far from its narrow, presumed original range.

2. Materials and methods

2.1. Sampling

A total of 107 individuals from 53 representative localities throughout the range of *Quasipaa boulengeri* and *Q. robertingeri* were included as ingroup members (Fig. 1). Among them, there were 58 sexed samples, 27 males and 31 females. The others were mostly tadpoles. Among the sexed samples, 31 individuals had been previously karyotyped, including some reciprocal translocation carriers (Qing et al., 2012; Xia et al., 2020). Based on the current phylogenetic hypothesis for *Quasipaa* (Yan et al., 2021), the species *Q. exilispinosa*, with four individuals from three sites, was used as outgroup when necessary. Altogether 111 samples were subjected to RAD-seq, including the aforementioned 32 samples for a preliminary analysis. Details of samples are provided in Table S1.

2.2. RAD-seq and data processing

Liver, muscle, or tail clips preserved in 95% ethanol were sent to Novogene (Beijing, China) for genomic DNA extraction, library construction, and sequencing. The library was prepared following the SLAF-seq approach (Sun et al., 2013) with a target insert size range of 215–240 bp. The restriction enzyme *MseI* was used in the restriction-ligation reaction, and a subset of the product was excluded by being cut by the restriction enzyme *HaeIII*. Enriched libraries were sequenced on an Illumina Novaseq 6000 platform in PE150 mode.

The quality of raw reads was checked using FastQC version 0.11.9 (Andrews, 2010). Exclusion of reads containing adapter sequences was conducted by Novogene. Further data cleaning was performed using the `process_radtags` program in Stacks version 2.60 (Catchen et al., 2013), removing reads with an average Phred33 score of < 10 within a 22-bp sliding window or an uncalled base.

We conducted a de novo assembly of loci using ipyrad version 0.9.87 (Eaton & Overcast, 2020), which applied reverse-complement clustering to our single-enzyme RAD-seq reads. To complete the assembly in a reasonable time, the clean reads were trimmed and filtered to avoid overloading with single-copy reads according to the ipyrad documentation (<https://ipyrad.readthedocs.io/>). Single-copy reads can result from sequencing errors that are more frequent toward the 3' end of reads. The last five bases of each read were trimmed using Trimmomatic version 0.39 (Bolger et al., 2014). Then, for each sample, a file containing only single-copy reads was obtained. This was achieved by sequentially using the `dedupe.sh` script in BBMap version 38.87 (Bushnell, 2014), the 'common' option in SeqKit version 2.3.0 (Shen et al., 2016), and the `filterbyname.sh` script in BBMap. According to this file, single-copy reads were removed by `filterbyname.sh`. In the ipyrad

pipeline, 3'-end bases with Phred33 scores below 20 were trimmed and three mitochondrial genomes were used as reference for excluding potential mitochondrial reads. They were under GenBank accession numbers KC686711 (*Q. boulengeri*; Shan et al., 2014 [DOI](#)), MZ895125 (*Q. robertingeri*; direct submission by Wang B and Shu G), and MT561179 (*Q. exilispinosa*; Wu et al., 2020 [DOI](#)). A minimum depth of 6 for both statistical and majority-rule base calling, a maximum cluster depth of 200 within samples, and an up to 70% occurrence for a shared heterozygous site were allowed. In a preliminary analysis of 22 representative ingroup samples, results obtained with clustering threshold values from 0.74 to 0.95 were compared. This parameter was determined as 0.89 according to the average observed heterozygosity, Pearson coefficient of correlation between missingness and genetic distance, and cumulative variance of the first five principal components (Ilut et al., 2014 [DOI](#); Mastretta-Yanes et al., 2015 [DOI](#); McCartney-Melstad et al., 2019 [DOI](#)). The correlation coefficient and cumulative variance were calculated using the `vcfMissingness.pl` and `vcfToPCAvrExplained.R` scripts (McCartney-Melstad et al., 2019 [DOI](#)), respectively. The maximum fraction of heterozygous bases in a consensus sequence was set to 0.11, i.e., one minus the clustering threshold as in the preliminary analysis for possible F₁-hybrid types.

The ipyrad SNP output was filtered using VCFtools version 0.1.17 (Danecek et al., 2011 [DOI](#)) by retaining genotypes with read depth ≥ 6 and SNPs present in $\geq 60\%$ of ingroup members. To minimize linkage disequilibrium, the `randSnps.pl` script (<https://www.biostars.org/p/313701/> [DOI](#)) was used to extract one random SNP per locus, resulting in the final main dataset. In addition, a dataset containing mainly pure *Q. robertingeri* samples was generated for population tree reconstruction. Two *Q. boulengeri* samples from one locality were included for rooting purposes. The ipyrad output was filtered with VCFtools by keeping genotypes with read depth ≥ 6 and SNPs present in all pure *Q. robertingeri* samples. Then, one SNP was randomly selected for each locus by `randSnps.pl`.

For fixed heterozygosity and haplotype network analyses, we also used Stacks to perform loci assembly of ingroup members. Stacks provides the observed heterozygosity for each SNP for each sample group and infers haplotype sequences using an algorithm relying on the co-observation of alleles in a read or read pair (Rochette et al., 2019 [DOI](#)). Currently, ipyrad provides unphased alleles as stated in its documentation. The `denovo_map.pl` pipeline of Stacks was applied to the clean reads, and a minimum of 3 reads was required to form a stack. Given the read length of 144 bp, the aforementioned ipyrad clustering threshold (0.89) suggested allowing 16 (i.e., $144 - 144 \times 0.89$) mismatches between stacks and between sample loci. These allowed mismatches were however both reduced to 12 for the reason of computational time based on a preliminary analysis. Loci present in $\leq 60\%$ of samples within a sample group were retained for subsequent analyses.

2.3. Individual heterozygosity, ancestry proportion, and fixed heterozygosity analyses

For each individual, the proportion of heterozygous SNPs was obtained from the main dataset and subsets of it using the `-het` option in VCFtools. For each SNP in this dataset, the first locus sequence in the `gphocs` output file from ipyrad was extracted with a custom script `gphocs1stSeq2fasta.pl` (<https://doi.org/10.57760/sciencedb.14822> [DOI](#)) (will be published on ScienceDB upon final acceptance of the manuscript; data private access link: <https://www.scidb.cn/en/s/URNIba> [DOI](#)) and TBtools version 1.120 (Chen et al., 2020 [DOI](#)). Using BLAST+ version 2.13.0 (Camacho et al., 2009 [DOI](#)) with an expectation value threshold of $1e-10$, these sequences were compared against the chromosome-level genomes of *Q. spinosa* (Hu, 2021 [DOI](#); Hu et al., 2022 [DOI](#)) and *Nanorana parkeri* (Genome Warehouse accession GWHBCKU000000000; Fu et al., 2022 [DOI](#)). *Nanorana* is closely related to *Quasipaa* (Pyron, 2014 [DOI](#)). Then loci with unique BLAST hits on homologous scaffolds of the two genomes were listed to obtain the proportion of heterozygous SNPs corresponding to individual and all 13 major scaffolds.

Ancestry proportions were inferred using ADMIXTURE version 1.3.0 (Alexander et al., 2009 [↗](#)) based on the main dataset, which had been converted into BED input files by PLINK version 1.9 (Chang et al., 2015 [↗](#)). For ingroup members, the number of ancestral populations (K) was set to 2. This value could be expected to result in only pure-species types and F_1 -hybrid types with ancestry proportions close to 50/50 when hybridogenesis occurred (e.g., Kimura-Kawaguchi et al., 2014 [↗](#); Dubey et al., 2019 [↗](#); Lavanchy et al., 2024 [↗](#)). To identify possible intraspecific hybrids, K values ranging from 1 to 10 were applied to a subset of samples in separate runs. Then the results from K values with minimum cross-validation errors were compared.

To examine the source of extraordinarily high individual heterozygosity, a simplified 12-individual dataset was extracted from the main dataset. It contained no missing data and three sample types according to ancestry proportions. In each type four samples were selected by having the most loci. Custom scripts were written to find SNPs being heterozygous for one type but showing alternatively fixed alleles, or the same allele, in the other two types.

For fixed heterozygosity analysis, given the ancestry proportion estimates, two Stacks sample groups with the same number of individuals were defined: F_1 -hybrid types and a randomly selected subset of pure *Q. boulengeri* types. If a retained Stacks locus was found to contain a SNP with an observed heterozygosity of 1 for a sample group, it was selected. Subsequently, using BLAST+ and an expectation value threshold of $1e-10$, its consensus sequence was compared against the aforementioned ipyrad sequences of loci with unique hits on homologous scaffolds.

2.4. Nuclear haplotype networks

Haplotype network construction was conducted using Stacks loci that matched an ipyrad locus with unique-hits on homologous scaffolds and exhibited fixed heterozygosity in F_1 -hybrid types. For each of the 13 major scaffolds, two loci also present in $\geq 60\%$ of pure-species types were randomly selected for analysis. Sequences with any undetermined bases were excluded. Statistical parsimony network analysis was performed using TCS version 1.21 (Clement et al., 2000 [↗](#)) at the 95% confidence level.

2.5. Mitochondrial DNA data and tree

Mitochondrial DNA sequences from three resources were combined to form a single dataset. First, mitochondrial sequences were extracted from the RAD-seq reads cleaned by process_radtags for ingroup members (Barth et al., 2020 [↗](#); Laczkó et al., 2022 [↗](#)). MITObim version 1.9.1 (Hahn et al., 2013 [↗](#)) was used to assemble mitochondrial sequences, with the aforementioned mitochondrial genomes of *Q. boulengeri* and *Q. robertingeri* as references. Reads were mapped to assembled sequences with BWA version 0.7.17 (Li & Durbin, 2009 [↗](#)) to obtain the read depth of each site using SAMtools version 1.9 (Li et al., 2009 [↗](#)). Sites with read depth < 6 were masked using BEDTools version 2.25.0 (Quinlan & Hall, 2010 [↗](#)). Second, four mitochondrial DNA fragments, *atp6-cox3*, *nad4*, *nad5*, and *cytb*, were PCR amplified and Sanger sequenced (ABI 3730) with primers in Table S2 [↗](#). For a subset of 59 samples, in most cases, all four fragments were sequenced. Lastly, published sequences of the *cytb* and *cox1* genes were retrieved from GenBank for 24 and 35 samples, respectively (Qing et al., 2012 [↗](#); Xia et al., 2016 [↗](#); Cao et al., 2018 [↗](#); Yuan et al., 2018 [↗](#)) (Table S1 [↗](#)). We also used the published mitochondrial genome of *Q. verrucospinosa* (KF199147; Zhang et al., 2018 [↗](#)) and the aforementioned mitochondrial genome of *Q. exilispinosa* as outgroups according to current mitochondrial genome tree reconstructions (Zhang et al., 2018 [↗](#)). The mitochondrial genome of *Q. boulengeri* has been found to exhibit tandem duplication and random loss of *trnW*, *trnA*, *trnN*, light-strand origin of replication, *trnC*, and *trnY* (Xia et al., 2016 [↗](#)). It may also be the case for *Q. robertingeri* (Zhang et al., 2018 [↗](#)). This “WANCY” region was not included. Alignment was performed using ClustalX version 2.1 (Larkin et al., 2007 [↗](#)) and checked by eye. Regions with alignment gaps in the ingroup were excluded.

A maximum likelihood tree was inferred from the combined dataset using RAxML version 8.2.12 (Stamatakis, 2014) with 1000 inferences and 2000 non-parametric bootstrap replicates. The rapid hillclimbing algorithm and automatic determination of the initial rearrangement setting were used. The bestfitting combination of data partitions and substitution models was determined using PartitionFinder version 2.1.1 (Lanfear et al., 2017) under the corrected Akaike information criterion. To mitigate potential overparameterization in an analysis of close relatives, only seven data blocks were defined for PartitionFinder evaluation. They were tRNA genes, 12S rRNA gene, 16S rRNA gene, the first, second, and third codon positions of protein-coding genes, and the *D-loop* region. The GTR+I+G substitution model was not evaluated due to the partial overlapping of the invariable sites parameter I and the among-site rate variation parameter G.

2.6. Principal coordinate plot, neighbor-joining tree, and population tree

To further assess genetic structure, using the main dataset and PLINK, an identity-by-state distance matrix between individuals was calculated for principal coordinates and neighbor-joining cluster analyses. Principal coordinates analysis was performed using GenALEx version 6.502 (Peakall & Smouse, 2012) for ingroup members. A neighbor-joining tree was constructed with 1000 bootstrap replicates using the Neighbor program in PHYLIP version 3.695 (Felsenstein, 2013). The script VCFbootstrap.pl (Zheng et al., 2020) was used to generate the 1000 bootstrap datasets.

To further examine relationships among pure *Q. robertingeri* samples and especially between those within and outside of the known species range, a population-tree approach was adopted. A maximum likelihood inference of population splitting and mixing was conducted using SNP allele frequencies and TreeMix version 1.13 (Pickrell & Pritchard, 2012). The allele frequencies for each sampling locality were calculated using PLINK. In separate TreeMix analyses, models allowing zero to six migration events were explored, and the results were compared.

3. Results

3.1. Individual heterozygosity, ancestry proportion, and fixed heterozygosity

The ipyrad genome-wide main dataset contained 45,523 SNPs, 8740–42,842 (mean = 30,600; 67.2%) for ingroup members and 2679–13,244 (8488) for outgroup individuals. Some individuals had much higher heterozygosities than the others, and the two forms occurred together and separately in the samples. The low- and high-heterozygosity samples were inferred as pure-species types and F_1 -hybrid types with approximately 50/50 ancestry proportions, respectively (Fig. 2). From 21 sites mainly around the Sichuan Basin, 40 F_1 types showed similar proportions of heterozygous loci with a mean of 0.167 and SD 0.004. For the other 71 samples including outgroups, a mean value of 0.029 and SD 0.015 were estimated, with a range from 0.009 found in site 30 to 0.081 for sample 28B (Figs. 1 and 2). Sexed F_1 types came from 13 sites, 18 females from 9 sites and 10 males from 7 sites. The interchromosomal reciprocal translocation carriers were F_1 types, and F_1 -type tadpoles had distinct, weak, or no transverse dark markings on dorsal tail (Table S1). For the 55 pure *Q. boulengeri* types, the smallest cross-validation errors were obtained with ADMIXTURE K priors 2 to 4. Sample 28B was the only one that appeared to be an intraspecific hybrid in all three cases (Fig. S1). The samples showing the lowest heterozygosities, 30B and 30D, were inferred to be pure *Q. robertingeri* types, the other ten of which were found in the known species range (Fig. 1).

In the main dataset, 20,912 SNPs were extracted from loci with unique BLAST hits on homologous major scaffolds of the *Q. spinosa* and *N. parkeri* genomes. These comprised 2379–3467 loci for the five larger scaffolds and 549–1219 loci for the eight smaller ones. Based on these SNPs, the above

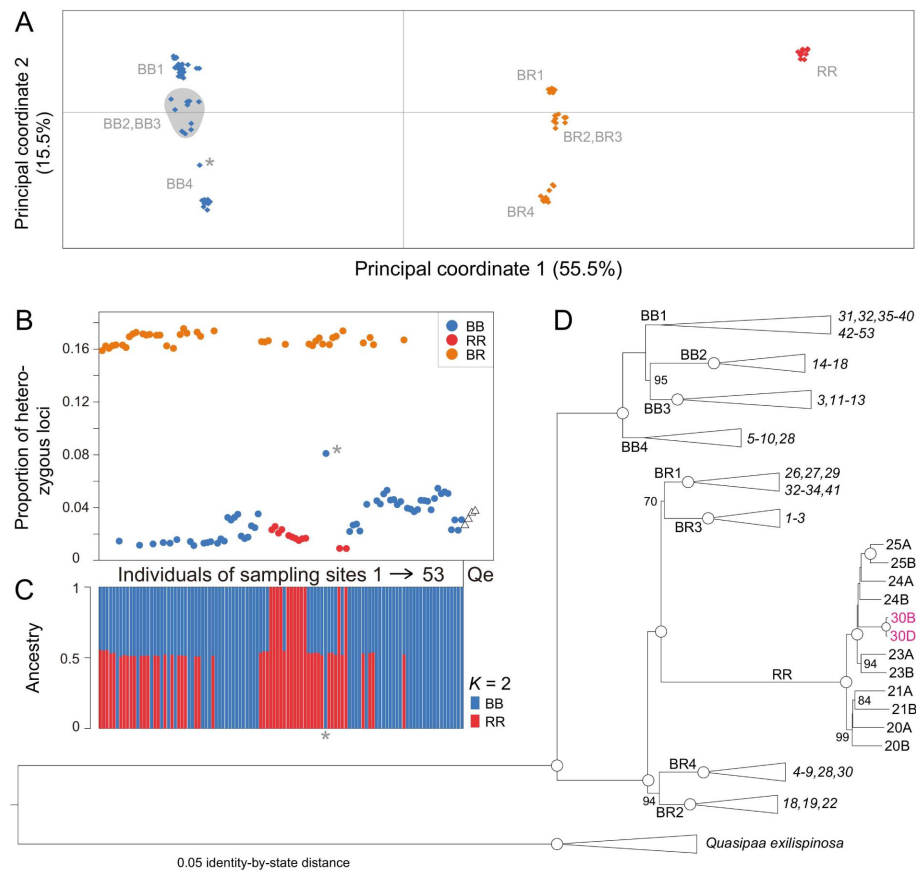


Figure 2

Principal coordinate plot (A), heterozygosity pattern (B), ADMIXTURE ancestry proportions (C), and neighbor-joining tree (D) based on the 45,523-SNP main dataset.

Italic numbers on the neighborjoining tree indicate sampling sites in [Fig. 1](#). Numbers beside nodes are bootstrap support values ≥ 70 , and circles on nodes indicate support values of 100. The sample 28B, a putative hybrid of different *Quasipaa boulengeri* lineages, is marked by an asterisk. In the neighbor-joining tree, the two individuals showing *Q. robertingeri* genotypes found far from the known species range are highlighted in purple. BB: *Q. boulengeri*; RR: *Q. robertingeri*; BR: putative *Q. boulengeri* $\times$ *Q. robertingeri* hybridogen; Qe: four *Q. exilispinosa* individuals.

distribution pattern of heterozygosity among samples remained unchanged for all and individual scaffolds (**Table S1** [↗](#)). Besides, two datasets containing at least 70% or 80% ingroup members at each locus were generated. Compared with the main dataset, they showed similar heterozygosity, ancestry proportions, principal coordinate plot, and neighbor-joining clustering results. No further analyses were applied to the two datasets. For clarity, a summary of most datasets used in this study is shown in **Table 2** [↗](#).

The 100% complete, 12-individual subset of the main dataset contained 25,046 SNPs. A total of 2567 sites were found to be heterozygous in the F_1 type and alternatively fixed in the two pure-species types, constituting the majority of the high heterozygosity estimates. The same pattern was observed when only SNPs from loci with unique BLAST hits were considered (**Table 3** [↗](#)). No sites were found to be heterozygous in the F_1 type while showing the same allele in both pure-species types.

A considerable amount of fixed heterozygosity of the F_1 type was detected. A total of 1260 Stacks loci exhibiting fixed heterozygosity among the 40 F_1 types ($\geq 60\%$ completeness) were found to have a BLAST hit to the aforementioned 20,912 unique-hit ipyrad loci. These comprised 156–210 and 24–66 loci corresponding to the larger and smaller scaffolds, respectively. No such fixed heterozygous loci were detected among the 40 randomly selected pure *Q. boulengeri* types.

3.2. Nuclear haplotype networks and sharing

For each of the 26 representative Stacks loci for fixed heterozygosity, a single parsimony network was constructed (**Figs. 3** [↗](#) and **S2** [↗](#)). Typically more haplotypes were found for the pure *Q. boulengeri* type (mean = 6.7) than for the pure *Q. robertingeri* type (1.6). Haplotypes were rarely shared between the two types, involving only three loci and three haplotypes. On the other hand, an F_1 type almost always shared one haplotype with the pure *Q. boulengeri* type and the other with the pure *Q. robertingeri* type. In the few exceptional cases between F_1 and *Q. robertingeri* types, a single substitution instead of haplotype sharing was observed. This allowed a ready identification of putative *Q. robertingeri* haplotypes.

Compared with the pure *Q. robertingeri* type, the F_1 type usually showed the same or fewer putative *Q. robertingeri* haplotypes for each locus. For all 26 loci, the former ($\bar{n} = 9$) and latter (26) respectively presented 42 and 34 such haplotypes and shared 30 of them. Besides, the haplotypes of the two pure *Q. robertingeri* types in site 30 were also found both in pure *Q. robertingeri* types from the known range and in F_1 types.

Haplotype sharing between F_1 and pure-species types in the same locality was common across the 26 loci. For the 21 F_1 types co-occurring with pure *Q. boulengeri* ones at nine sites, it was observed in 94.4% of the 357 cases with available data. For the two F_1 types sampled with pure *Q. robertingeri* ones at site 30, it was found in 42 of 43 cases. Of the 26 loci, 21 had at least five haplotypes found in pure *Q. boulengeri* ones. Besides, six F_1 -type-sample only sites had a nearby pure *Q. boulengeri* sample, i.e., sites 1, 2, 4, 33, 34, and 41 (**Fig. 1** [↗](#)). When these sites were also considered, local haplotype sharing between F_1 and pure *Q. boulengeri* types was observed in 92.8% of the 347 cases across the 21 loci.

3.3. Mitochondrial DNA data and tree

Mitochondrial sequence fragments were assembled from RAD-seq reads for all ingroup members, from 2472 to 6983 bp per sample with an average of 5099 bp. A total of 179 newly obtained Sanger sequences were deposited under GenBank accessions PP065924–36 and PP083013–178 (**Table S1** [↗](#)). The combined mitochondrial dataset contained 10,568 sites and was 52.7% complete. For ingroup members, 1514 sites were variable and 1154 sites were potentially parsimony-informative. This dataset is available on ScienceDB (<https://doi.org/10.57760/sciencedb.14822> [↗](#))

Dataset feature	Nuclear genotype	Number of SNPs/sites (1 random SNP per locus)	Completeness	Analysis applied
The main dataset	55 BB, 40 BR, 12 RR, outgroup	45,523 SNPs	67.2% (ingroup)	Genome-wide proportion of heterozygous SNPs, principal coordinates analysis, neighbor-joining clustering, maximum likelihood estimation of ancestry proportions (ingroup, 2 ancestral populations)
Unique hit	55 BB, 40 BR, 12 RR	20,912 SNPs	67.1%	Genome-wide and chromosome-level proportions of heterozygous SNPs
12 individuals	4 BB, 4 BR, 4 RR	25,046 SNPs	100%	Genome-wide proportions of heterozygous SNPs (SNPs being heterozygous for BR while showing one allele in BB and the other in RR considered or excluded)
12 individuals and unique hit	4 BB, 4 BR, 4 RR	12,183 SNPs	100%	Genome-wide proportions of heterozygous SNPs (SNPs being heterozygous for BR while showing one allele in BB and the other in RR considered or excluded)
<i>Q. boulengeri</i>	55 BB	45,523 SNPs	66.0%	Maximum likelihood estimation of ancestry proportions (to identify possible intraspecific hybrids in BB)
<i>Q. robertingeri</i>	12 RR, 2 BB	8535 SNPs	100% (RR)	Maximum likelihood inference of population splitting and mixing (to further examine the insular occurrence in RR)
26 (presumably 2 per chromosome) of 1260 loci with fixed heterozygosity in BR and unique hit	38 BB, 26 BR, 9 RR (means)	145 sites (mean)	100%	Statistical parsimony haplotype network for each locus
Mitochondrial DNA	55 BB, 40 BR, 12 RR, outgroup	10,568 sites	52.7%	Maximum likelihood gene tree

Table 2

Dataset summaries.

Unique hit: loci with unique BLAST hits on homologous chromosome-level scaffolds of two genomes of the related *Quasipaa spinosa* and *Nanorana parkeri*; BB: *Q. boulengeri*; RR: *Q. robertingeri*; BR: putative *Q. boulengeri* × *Q. robertingeri* hybridogen.

Cluster	ID	25,046-site data		12,183-site data	
		H	H'	H	H'
BB	5D	243	243	114	114
	9D	300	300	136	136
	9E	317	317	148	148
	9F	314	314	147	147
RR	21B	538	538	270	270
	23A	477	477	217	217
	30B	205	205	99	99
	30D	207	207	102	102
BR	8B	4,412	1,845	1,823	804
	9G	4,367	1,800	1,812	793
	27B	4,194	1,627	1,756	737
	30C	4,411	1,844	1,808	789

Table 3

Heterozygous SNP numbers of twelve individuals in subsets of the 45,523-SNP main dataset.

The 25,046-site subset contained no missing data and 12,183 sites each corresponding to a locus with unique BLAST hits on homologous scaffolds of two published genomes of *Quasipaa spinosa* and *Nanorana parkeri*. ID: individual ID; BB: *Q. boulengeri*; RR: *Q. robertingeri*; BR: putative BB x RR hybridogen. SNPs being heterozygous for BR while showing one allele in BB and the other in RR were considered (H) or excluded (H').

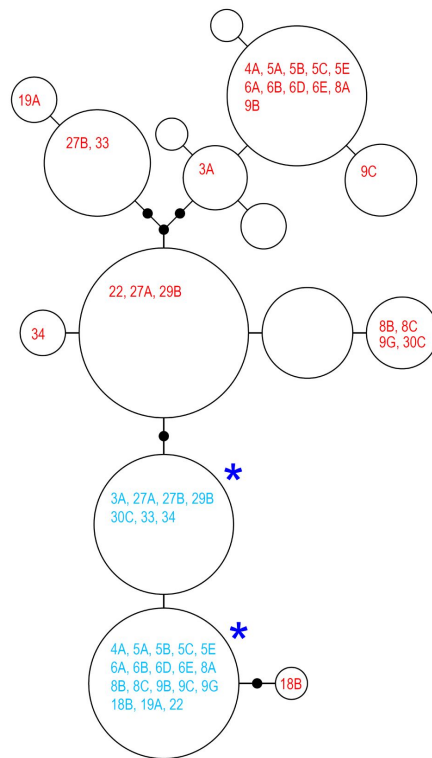


Figure 3

Statistical parsimony network for haplotypes of a single exemplary locus, CLocus_761519 of 144 bp, from individuals of *Quasipaa boulengeri*, *Q. robertingeri*, and putative *Q. boulengeri* × *Q. robertingeri* hybridogen genotypes.

This locus was heterozygous in putative hybridogens and had unique BLAST hits on the first scaffolds of published *Q. spinosa* and *Nanorana parkeri* genomes. Lines represent single substitutions. Filled circles are haplotypes not encountered. The other circles' sizes are proportional to haplotype frequency. Putative hybridogens' sample IDs are shown. Samples of *Q. robertingeri* exhibited haplotypes marked by asterisks.

(will be published upon final acceptance of the manuscript; data private access link: <https://www.scidb.cn/en/s/URnIba>). For the RAxML analysis, a six-partition scheme with a GTR+G model for each partition was selected: tRNA, rRNA, *D-loop*, and each of the three codon positions.

F₁ types exhibited mitochondrial haplotypes clustering together with those of co-occurring and nearby pure *Q. boulengeri* types on the generally well-supported RAxML tree (**Fig. 4**) and often shared haplotypes with the latter. In the *Q. robertingeri* clade Qr, haplotypes from the eastern and western known distribution range were grouped separately. In the other clade Qb, haplotypes clustered into four subclades.

Mitochondrial haplotypes from *Q. boulengeri* were detected in three pure *Q. robertingeri* types. Sample 21A had a haplotype closely related to that of a nearby F₁ type, the others, at site 30, shared haplotypes with co-occurring F₁ or nearby pure *Q. boulengeri* types. No mitochondrial haplotypes from *Q. robertingeri* were found in pure *Q. boulengeri* types.

3.4. Principal coordinate plot, neighbor-joining tree, and population tree

The principal coordinate plot showed three discrete clusters corresponding to the F₁ and two pure-species types, with the former located approximately intermediate between the latter two (**Fig. 2**). Among them, 71.1% of the observed genetic variation was explained by the first two principal coordinates. Much more variation was observed within the F₁ or pure *Q. boulengeri* types than in the pure *Q. robertingeri* type. Subclusters could be identified in the former two.

The neighbor-joining tree was generally well supported. Two pure-species clusters with bootstrap support of 100 were observed. The two pure *Q. robertingeri* types from site 30 were grouped with those from the eastern known species range (sites 23–25 in **Fig. 1**) with strong support. For the F₁ type, subclusters BR1, BR2, BR3, and BR4 were identified, corresponding to the four subclusters of the pure *Q. boulengeri* type, BB1, BB2, BB3, and BB4, respectively (**Fig. 2D**). When applying these subclusters to the principal coordinate plot, parallel structures were readily detected. Along the second principal coordinate, a BB2+BB3 cluster was located between BB1 and BB4, and BR2+BR3 was located between BR1 and BR4. Besides, BB1–4 and BR1–4 corresponded to the four subclades of the mitochondrial clade Qb (**Fig. 4**). The full neighbor-joining tree is available as **Fig. S3**.

The dataset for TreeMix analysis of pure *Q. robertingeri* types contained 8535 SNPs. Three migration edges between populations from the known species range were found to increase the likelihood significantly (all $P < 0.01$). With these edges added or not, the same maximum likelihood tree topology was obtained in which site 30 was again nested within the eastern clade (**Fig. 4**).

4. Discussion

4.1. Hybridogenesis and insular occurrence

In the *Quasipaa boulengeri*-*Q. robertingeri* complex, hybridogenesis can be inferred and distinguished from other systems where hybrids are F₁ types. Our dataset consisted of two low-heterozygosity parental types and one hybrid type characterized by much higher genome-wide heterozygosity (**Fig. 2B**), ~50/50 ancestry proportions (**Fig. 2C**), and a large number of fixed heterozygous loci (6%) throughout the genome. Previous karyotyping of these frogs has identified solely $2n = 26$ diploid individuals (e.g., Tan & Wu, 1987; Xia et al., 2020), which when genotyped appear as either F₁ or parental types. With F₁ type hybrids widely detected (**Fig. 1**), the known natural range of *Q. robertingeri* is narrow (e.g., Qing et al., 2012; Yan et al., 2013; Fei, 2020; Wang et al., 2021). Hybrid sterility, or any mode requiring many local hybrid

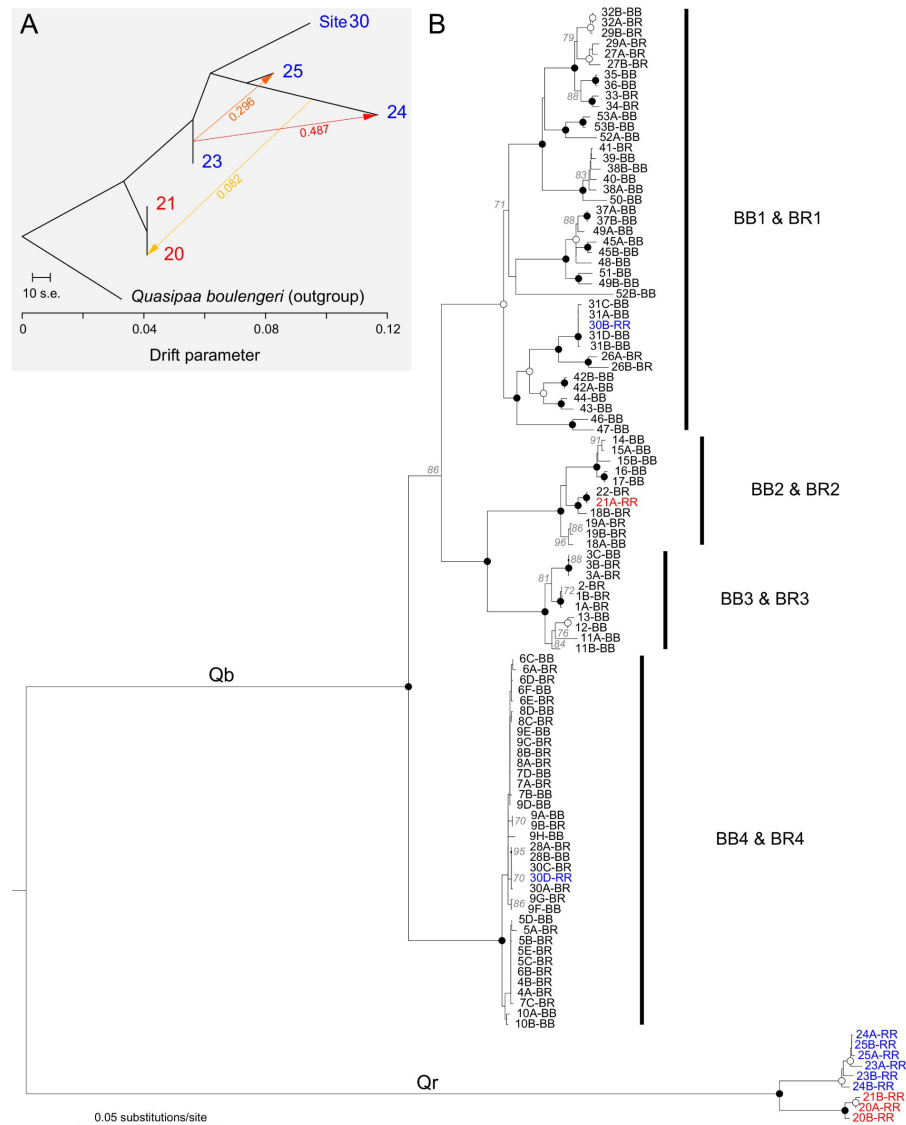


Figure 4

Maximum likelihood population tree of *Quasipaa robertingeri* with three migration edges (arrows) inferred using TreeMix (A) and RAXML mitochondrial gene tree (B).

The TreeMix tree was based on the 8,535-SNP nuclear dataset of 100% completeness for ingroups. The number below an arrow is migration weight that represents the fraction of ancestry derived from the migration edge. s.e.: average standard error of the sample covariance matrix of allele frequencies. The RAXML tree was based on a 10,568-site dataset of 52.7% completeness and six partitions. Italic numbers beside nodes are bootstrap support values ≥ 70 , and circles on nodes indicate support values of 100 (closed) and 98–99 (open). The characters BB, RR, or BR following the sample ID indicate *Q. boulengeri*, *Q. robertingeri*, or putative hybridogen nuclear genotypes, respectively. Lineage names within the Qb clade correspond to cluster names in [Fig. 2](#). Outgroups are not shown for the RAXML tree.

origins, can be ruled out by such a lack of coexistence of hybrids with both parental species (**Table 1**). In the present study, 10 of the 27 sexed F_1 type adults were males from 7 sites (**Table S1**). This is not expected for parthenogenesis, gynogenesis, or kleptogenesis. In lineages exhibiting such a mode, males are absent or very rare (Avisé, 2008; Tilquin & Kokko, 2016; Fujita et al., 2020). For 17 sites across the F_1 type range, local nuclear haplotype sharing between F_1 and pure *Q. boulengeri* types occurred at a rate of 92.8% in 21 representative loci. This is unexpected for androgenesis, parthenogenesis, gynogenesis, and kleptogenesis when assuming a single or few local hybrid origins. In these modes a codistributed parental species has only occasional (kleptogenesis) or no contribution to the nuclear genome (Lampert & Schartl, 2010; Lehtonen et al., 2013; Morgado-Santos et al., 2017). Furthermore, the widespread high mitochondrial DNA similarity between co-distributed F_1 and *Q. boulengeri* types (**Fig. 4B**) is not expected for parthenogenesis, gynogenesis, and kleptogenesis of a single or few local hybrid origins (Bogart et al., 2009; Bi & Bogart, 2010; Schmidt et al., 2011). Last, the two pure *Q. robertingeri* types found with F_1 types at site 30 can hardly be explained by even a combination of non-hybridogenetic processes. With divergent non-conspecific mitochondrial genomes (Plotner et al., 2008; Liu et al., 2010), these two samples were nested deeply among those from the narrow known range approximately 500 kilometers away based on genome-wide nuclear data (**Figs. 2D** and **4**). It is unlikely that, through androgenesis, they have been produced by parthenogenetic hybrid females and local relict *Q. robertingeri* males (Tinti & Scali, 1996). They may also in theory have arisen from androgenesis with homologous haploid sperm formed by hybrids through a meiotic hybridogenesis-like process (Arai, 2023; Lafond & Angers, 2024). However, their low but apparent and similar chromosome-level heterozygosities indicate other infrequent events, polyspermy involving different males (Talevi, 1989; Schwander & Oldroyd, 2016; Hater et al., 2020; Iwao & Ueno, 2022). In contrast, all the above findings are consistent with a hybridogenetic system in which hybridogens can share nuclear and mitochondrial genetic variation with local parental species and can produce heterozygous pure-species offspring (Vorburger, 2001a, b; Dufresnes & Mazepa, 2020; Doležálová-Kašánková et al., 2021).

Along with the clustering of pure *Q. robertingeri* types, two lines of evidence also support an “out of the known range” origin for the widespread *Q. robertingeri* nuclear genomes. First, the mitochondrial lineage corresponding to this species is endemic to the narrow known range, and it shows a significant east-west division that suggests long-term persistence (**Figs. 1** and **4B**) (Yan et al., 2013; Wang et al., 2021). Second, for nuclear loci, the variation detected within the known range is greater than and overlaps substantially with that found outside it. For the 26 representative loci, on average 7 pure-species and 28 F_1 /pure-species types were respectively sampled within and outside the known range. However, 23.5% more putative *Q. robertingeri* haplotypes were detected in the former, with 88.2% such haplotypes observed in the latter shared by the former. Based on the 45,523-SNP data, parallel geographic structures were inferred for pure *Q. boulengeri* and F_1 types (**Figs. 1** and **2A**). This also agrees with the “out of the known range” scenario. Because nuclear variation in *Q. boulengeri* was much higher than that of *Q. robertingeri* from the known range, among hybridogens the former could contribute dominantly in characterizing geographic structure.

In addition, it is unlikely that accidental transport of tadpoles (along the Yangtze River) or human-mediated translocation from the known range has caused the insular occurrence of *Q. robertingeri* nuclear genomes at site 30 (**Fig. 1**). In both cases, the F_1 types of this site are expected to have a local *Q. boulengeri* haploid nuclear genome and hence belong to subcluster BR1. However, they were nested in subcluster BR4 and were most closely related to the F_1 type from an intermediate site (28) between the known range and their locality (**Fig. S3**). Moreover, three of the four samples from site 30 also had mitochondrial haplotypes most closely related to the one found in site 28 (**Fig. 4B**).

To our knowledge, this study provides the first evidence consistent with a hypothesis that hybridogenesis mediates the occurrence of individuals far from the continuous distribution of conspecifics. Such an occurrence obviously does not guarantee the successful establishment of a *Q. robertingeri* population, but the possibility deserves further exploration in addition to cytogenetic examination of hybridogenesis. The reason is that if this kind of reconstituted population does exist, then the finding will introduce a new mode of range expansion and extend our understanding of the interaction between narrow-ranged and widespread relatives.

4.2. Individual heterozygosity and features of the spiny frog system

Atypical reproductive systems of hybrid origin are unusual and useful research resources (Laskowski et al., 2019). For instance, hybridogenesis suggests an approach to preserve a species in non-optimal conditions by taking advantage of a relative and of hybrid vigor (Dubey & Dufresnes, 2017). This study constitutes another example of heterozygosity patterns inspiring a test of atypical reproduction, with the discovery of a putative hybridogenetic complex unrelated to any known one for the first time since 2014 (Kimura-Kawaguchi et al., 2014; Lavanchy & Schwander, 2019; Dufresnes et al., 2024). From a dataset of genome-wide loci, individual heterozygosity estimates are among the first statistics that one can obtain. If some estimates are much higher than the others which is the case here, the pattern can serve as a clue to the presence of unusual reproductive modes (e.g., Jaron et al., 2021; Lavanchy et al., 2024). Such datasets are common in the era of high-throughput sequencing. However, a simple Google Scholar search has returned only 1850 or 812 articles dated between 2014 and June 3, 2025 with the phrase “individual heterozygosity/heterozygosities” or “proportion/proportions of heterozygous loci/sites”, respectively. It implies that this variable has not been reported in most cases, making the accumulating genome-wide data a resource for searching for individual heterozygosity clues of atypical reproduction of interspecific hybrid origin.

Compared with documented hybridogenetic systems, *Q. boulengeri* and *Q. robertingeri* occur in a relatively poorly studied region from a natural history perspective, East Asian mainland. The former have been reported from North America, Europe, Australia, and the Japanese islands (Suzuki et al., 2020; Majtánová et al., 2021). Research efforts are not equally distributed across taxa. From under-studied taxa, atypical reproduction may even be found as a by-product of efforts to understand the study system.

As revealed by conventional staining, the *Q. boulengeri* x *Q. robertingeri* system shows structurally abnormal, non-Y/W chromosomes which are regionally widespread. The fixation or partial fixation of such chromosomal mutations is generally rare probably due to deleterious effects and meiotic pairing abnormalities (Wallace et al., 1999; Bidau & Marti, 2010; Mezzasalma et al., 2017, 2021). In hybridogenesis, clonal genomes, which do not face meiotic pairing problems and recombination, are prone to accumulation of deleterious mutations and can exhibit diagnostic chromosomal rearrangements (Suzuki et al., 2017, 2020; Lavanchy & Schwander, 2019; Dufresnes & Mazepa, 2020). In the spiny frog system, female and male carriers of an interchromosomal reciprocal translocation (karyotype IRT) have been found in some populations at the western edge of the Sichuan Basin (Fig. 1). Furthermore, there is a tight linkage between the two involved chromosomes, with other co-occurring abnormal karyotypes rarely observed (Qing et al., 2012). In this region, our samples with normal karyotype can be either pure *Q. boulengeri* ($n = 10$) or putative hybridogens ($n = 4$). As predicted by a hypothesis that the linkage is maintained by clonal transmission of the *Q. robertingeri* haploid genome, the 6 samples of karyotype IRT all belong to the latter (chi-square $P < 0.01$). The rest few samples of rare abnormal karyotypes are also putative hybridogens showing chromosome-level heterozygosity estimates of similarly large values, implying a complicated scenario with for instance possible subsequent rearrangements. When non-sex-specific abnormal chromosomes are encountered, hybridogenesis may serve as a hypothesis to be tested.

Quasipaa boulengeri and *Q. robertingeri* show significant mitochondrial divergence comparable to those between parental species of known hybridogenetic systems. These parental species are expected to be similar enough to produce fertile offspring. They are also expected to attain a certain divergence before hybridization leads to a transition to hybridogenesis rather than to genetic admixture (Dufresnes & Mazepa, 2020 [DOI](#); Barley et al., 2022 [DOI](#)). Currently, genome-wide nuclear data for a comparison across known hybridogenetic systems are not available. However, a mitochondrial *cytb* partial sequence of 948 bp is accessible for most of these parental species in GenBank. By arbitrarily using one such sequence to represent a *Pelophylax* frog species (JN627421, 22, MF667576; Hofman et al., 2012 [DOI](#); Dufresnes et al., 2017 [DOI](#)), one can obtain uncorrected *p*-distances of 0.141 and 0.133. They are for the two parental species pairs *P. ridibundus*–*P. lessonae* and *P. ridibundus*–*P. perezi* (Dufresnes & Mazepa, 2020 [DOI](#)), respectively. The corresponding value is 0.113 between *Q. boulengeri* (KC686711) and *Q. robertingeri* (MZ895125). For hybridogenetic fishes, such *p*-distances can be estimated as 0.100 and 0.082 in *Hexagrammos* (AB290742, 48, 69; Kimura et al., 2007 [DOI](#)), 0.129–0.133 in *Poeciliopsis* (AF412139, 51, MK860197, 98, KX229692; Mateos et al., 2002 [DOI](#), 2019 [DOI](#); Jeon et al., 2016 [DOI](#)), and 0.098–0.103 in *Hypseleotris* (DQ468188, 236, 250, 296; Thacker et al., 2007 [DOI](#)). In *Squalius*, a mitochondrial lineage corresponding to one of the two parental species has not been found (Matos et al., 2019 [DOI](#)). In the other, *Bacillus* stick insect hybridogenetic system, a similar level of mitochondrial divergence can also be observed. An uncorrected *p*-distance of 0.074 is estimated for a 675-bp fragment of the *cox2* gene (AF148299, 315; Mantovani et al., 2001 [DOI](#)), which evolves at a rate close to that of the *cytb* gene (e.g., Hofman et al., 2012 [DOI](#)). These estimates are higher than many shallow interspecific distances (e.g., Young et al., 2013 [DOI](#); Tarvin et al., 2017 [DOI](#)), and their similarity agrees with a previous comparison (Stöck et al., 2021 [DOI](#)). This suggests an investigation of the range of nuclear divergence for various parental species pairs.

Data availability

Custom scripts used and the data underlying this article including RAD-seq reads, SNP datasets, BLAST query sequences, and sequence datasets of nuclear and mitochondrial loci are available on ScienceDB (<https://doi.org/10.57760/sciencedb.14822> [DOI](#)) (will be published upon final acceptance of the manuscript; data private access link: <https://www.scidb.cn/en/s/URnIba> [DOI](#)). Mitochondrial sequences obtained by the Sanger method were deposited in GenBank under accessions PP065924–36 and PP083013–178.

Supplementary Files

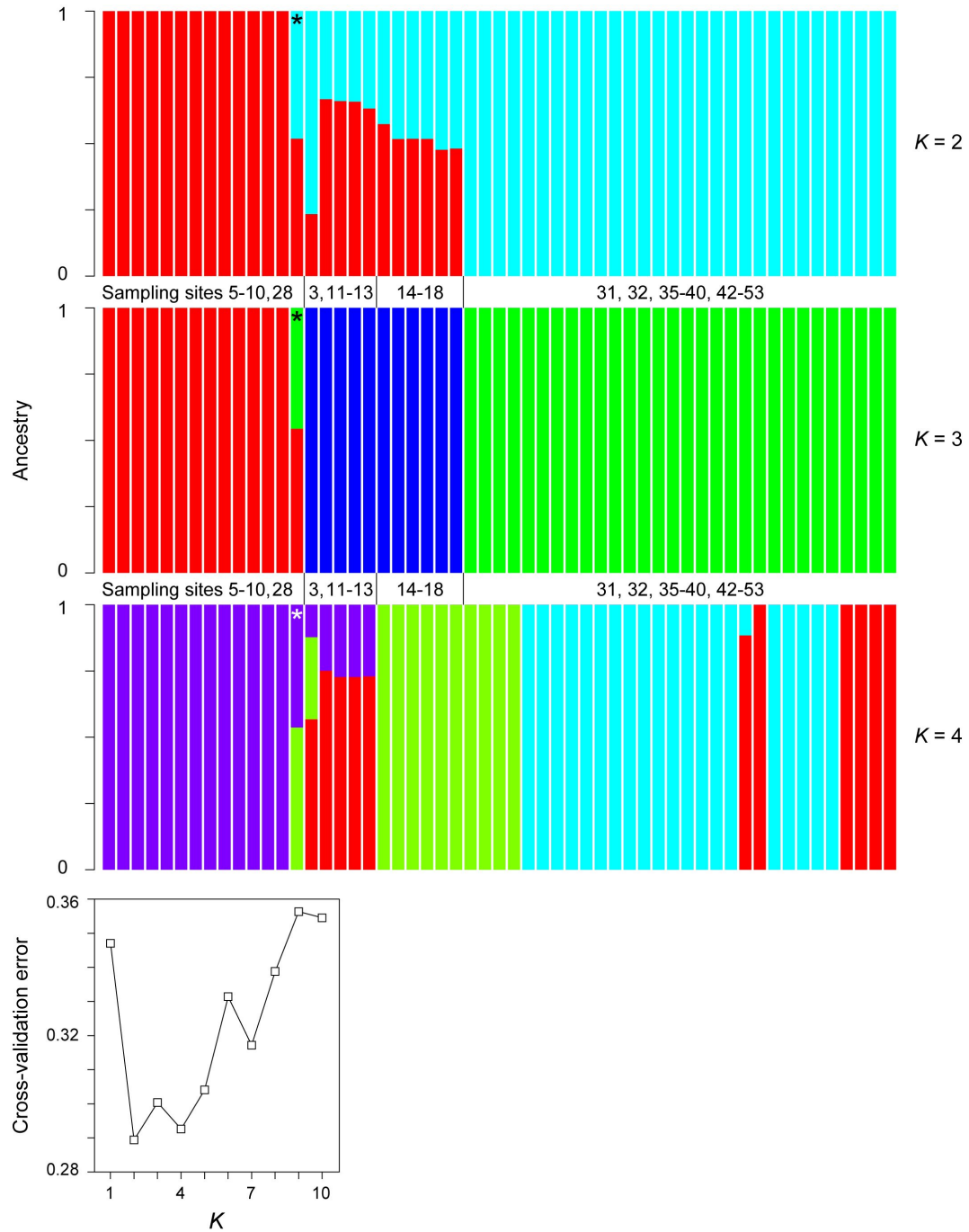
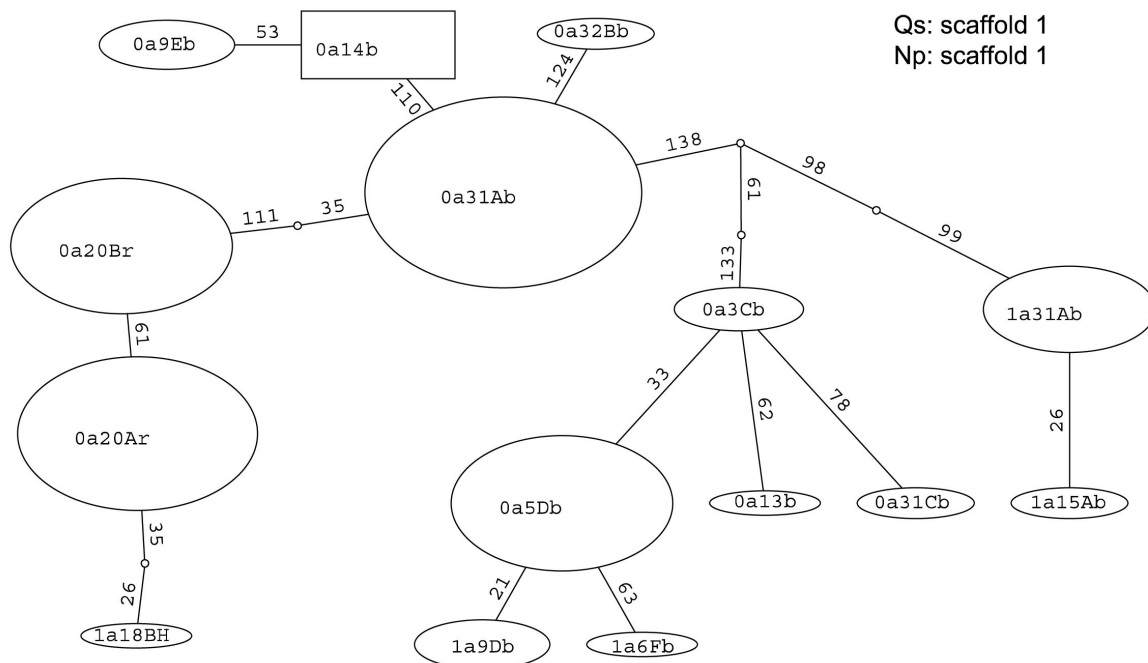


Figure S1.

ADMIXTURE cross-validation plot and ancestry proportions for 55 *Quasipaa boulengeri* individuals based on the 45,523-SNP dataset.

*: Sample 28B.



Haplotype list:

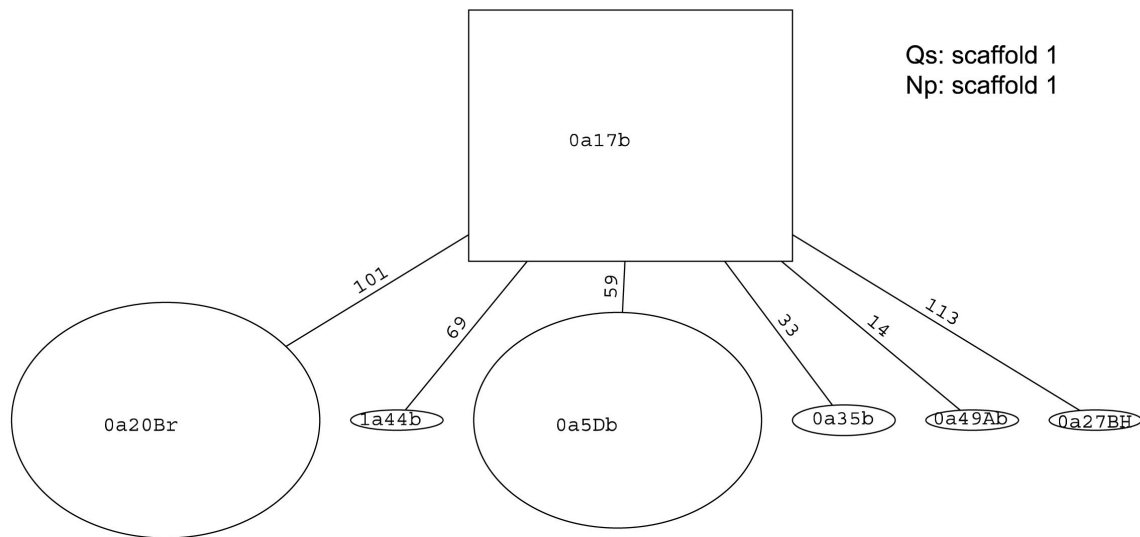
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0a5EH 0a6AH 0a6BH 0a6DH 0a6EH 0a8AH 0a9BH
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- 0a9Eb : 1a8BH 1a8CH 1a9GH 1a30CH
- 0a13b :
- 1a15Ab : 1a19AH
- 0a31Ab : 0a31Bb 0a31Db 0a36b 0a37Ab 0a39b 1a39b 0a45Ab 0a45Bb 1a45Bb 1a46b 1a47b
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0a53Bb 1a53Bb 1a22H 1a27AH 1a29BH
- 1a31Ab : 1a31Bb 1a31Db 1a32Bb 1a36b 1a37Ab 1a45Ab 1a48b 1a50b 1a27BH 1a33H
- 0a31Cb : 1a31Cb
- 0a32Bb : 1a34H
- 0a20Ar : 1a20Ar 0a21Ar 1a21Ar 1a4AH 1a5AH 1a5BH 1a5CH 1a5EH 1a6AH 1a6BH 1a6DH
1a6EH 1a8AH 0a8BH 0a8CH 1a9BH 0a9CH 0a9GH 0a18BH 0a19AH 0a22H
- 0a20Br : 1a20Br 0a23Ar 1a23Ar 0a24Ar 1a24Ar 0a25Ar 1a25Ar 0a25Br 1a25Br 0a30Dr 1a30Dr
0a3AH 0a27AH 0a27BH 0a29BH 0a30CH 0a33H 0a34H
- 1a18BH :

A

Figure S2.

Statistical parsimony networks for haplotypes of 26 loci.

Corresponding scaffolds of published genomes of *Quasipaa spinosa* (Qs) and *Nanorana parkeri* (Np) are shown. The 0a and 1a in the sequence name indicate allele 0 and allele 1, respectively. The last character in a sequence name, b, r, or H, indicates *Q. boulengeri*, *Q. robertingeri*, or putative hybridogenetic individuals, respectively. Between them, the sample ID is given. Lines represent single substitutions, and numbers beside lines are site numbers. Circles are haplotypes not encountered.

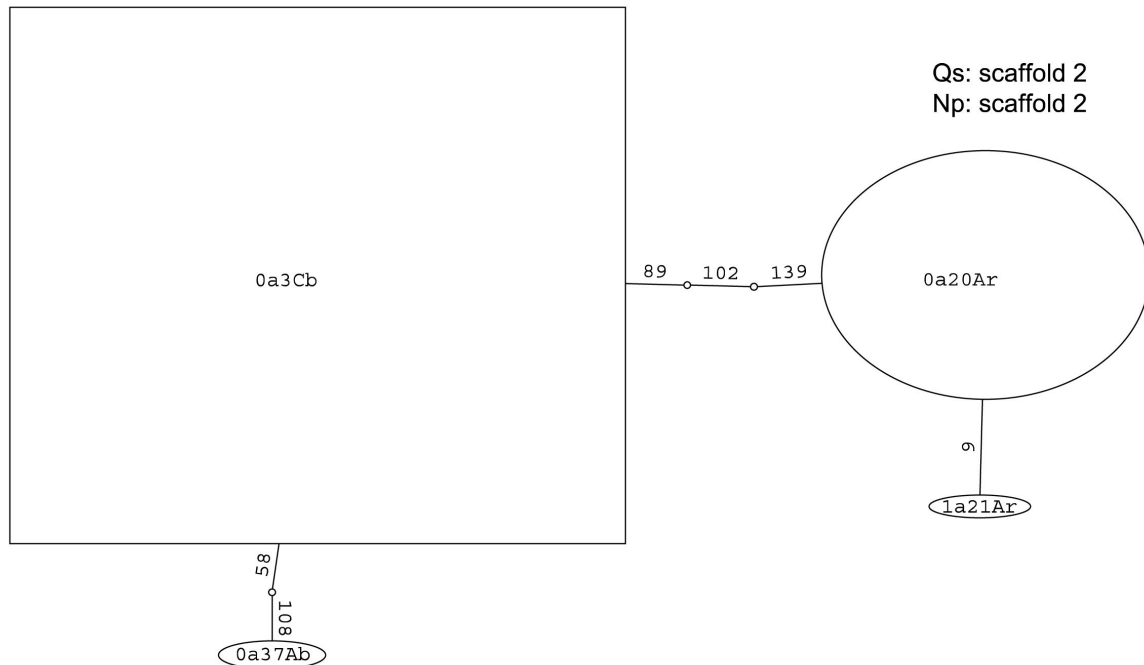


Haplotype list:

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          0a4BH 0a5AH 0a5BH 0a5CH 0a5EH 0a6AH 0a6BH 0a6EH 0a8AH 0a8BH 0a9BH
          0a9CH 0a9GH 0a28AH 0a30AH 0a30CH
- 0a17b : 1a17b 0a18Ab 1a18Ab 1a28Bb 0a31Ab 1a31Ab 0a31Bb 1a31Bb 0a31Cb 1a31Cb 0a31Db
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          0a34H
          0a27AH 0a32AH
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- 1a44b :
- 0a49Ab :
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B

Figure S2. (continued)

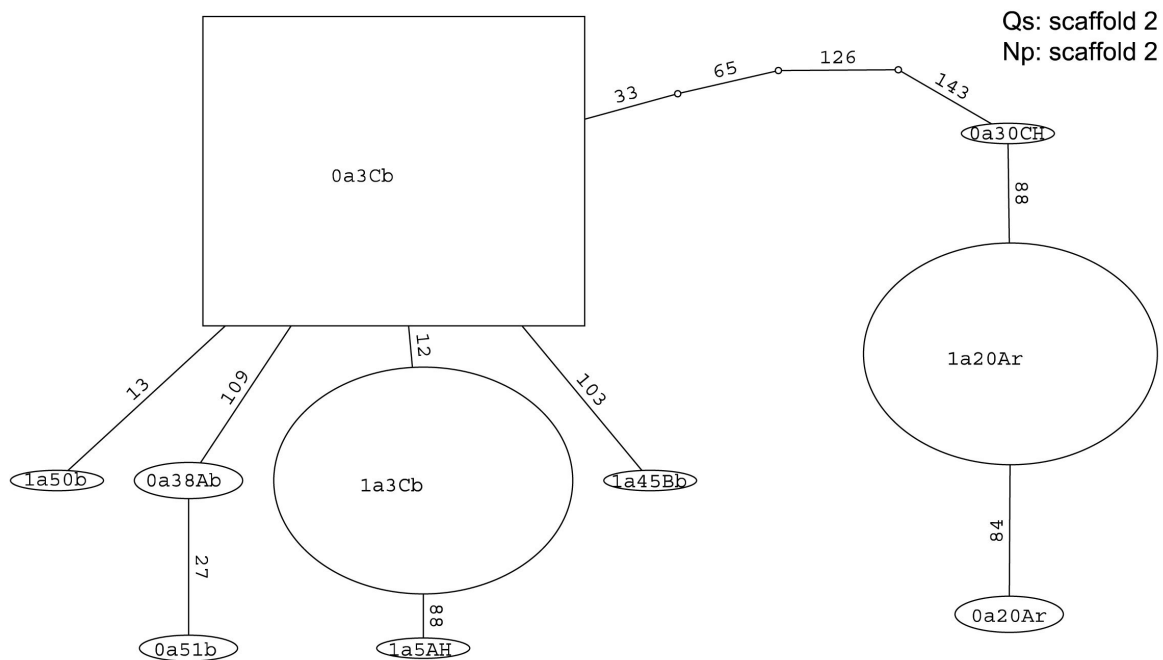


Haplotype list:

```
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          0a51b 1a51b 0a52Ab 1a52Ab 0a52Bb 1a52Bb 0a53Ab 1a53Ab 0a53Bb 1a53Bb 0a42Ab
          1a42Ab 1a3BH 1a4AH 1a4BH 1a5AH 1a5BH 1a5CH 1a5EH 1a6AH 1a6BH 1a6EH
          1a8AH 1a8BH 1a8CH 1a9BH 1a9CH 1a9GH 1a22H 1a27AH 1a27BH 1a29BH 1a30CH
          1a33H 1a34H 1a41H
- 0a37Ab : 0a37Bb
- 0a20Ar : 1a20Ar 0a20Br 1a20Br 0a21Ar 0a21Br 1a21Br 0a23Ar 1a23Ar 0a25Ar 1a25Ar 0a25Br
          1a25Br 0a30Br 1a30Br 0a30Dr 1a30Dr 0a3BH 0a4AH 0a4BH 0a5AH 0a5BH 0a5CH
          0a5EH 0a6AH 0a6BH 0a6EH 0a8AH 0a8BH 0a8CH 0a9BH 0a9CH 0a9GH 0a22H
          0a27AH 0a27BH 0a29BH 0a30CH 0a33H 0a34H 0a41H
- 1a21Ar :
```

C

Figure S2. (continued)

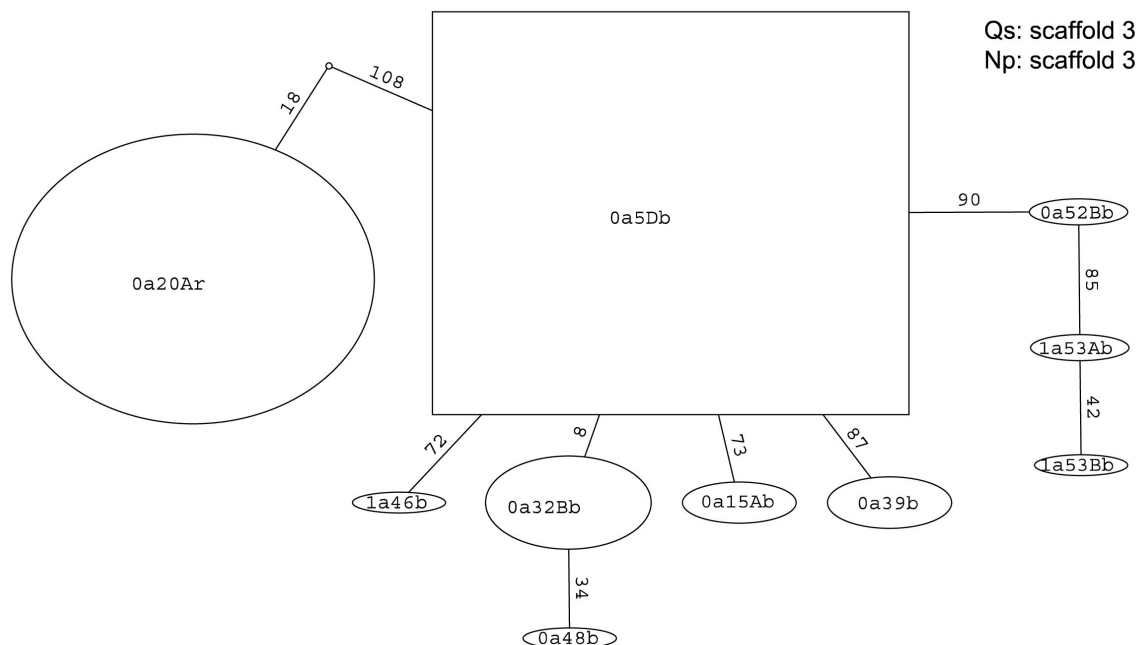


Haplotype list:

```
- 0a3Cb : 0a28Bb 0a31Ab 1a31Ab 0a31Bb 1a31Bb 0a31Cb 1a31Cb 0a31Db 1a31Db 0a32Bb
          1a32Bb 0a35b 1a35b 0a36b 1a36b 0a37Bb 1a37Bb 0a40b 1a40b 0a43b
          1a43b 0a44b 1a44b 0a45Ab 1a45Ab 0a45Bb 0a46b 1a46b 0a47b 1a47b
          0a48b 1a48b 0a49Ab 1a49Ab 0a49Bb 1a49Bb 0a50b 0a52Ab 1a52Ab 0a52Bb
          1a52Bb 0a53Ab 1a53Ab 0a53Bb 1a53Bb 0a42Ab 1a42Ab 1a2H 1a26AH 1a26BH
          1a27AH 1a27BH 1a29BH 1a41H 1a1AH 1a1BH
- 1a3Cb : 0a5Db 1a5Db 0a6Cb 1a6Cb 0a6Fb 1a6Fb 0a7Db 1a7Db 0a9Ab 1a9Ab
          0a9Db 1a9Db 0a9Eb 1a9Eb 0a9Fb 1a9Fb 0a9Hb 1a9Hb 0a10Ab 1a10Ab
          0a10Bb 1a10Bb 0a18Ab 1a18Ab 1a28Bb 1a3BH 1a4BH 1a5CH 1a6AH 1a6BH
          1a6DH 1a6EH 1a8BH 1a8CH 1a9BH 1a9CH 1a9GH 1a19BH 1a28AH 1a30CH
- 0a38Ab : 1a38Ab 0a38Bb 1a38Bb
- 1a45Bb :
- 1a50b :
- 0a51b : 1a51b
- 0a20Ar : 0a21Ar 0a21Br 1a21Br
- 1a20Ar : 0a20Br 1a20Br 1a21Ar 0a23Ar 1a23Ar 0a23Br 1a23Br 0a24Ar 1a24Ar 0a24Br
          1a24Br 0a30Br 1a30Br 0a30Dr 1a30Dr 0a2H 0a3BH 0a4BH 0a5AH 0a5CH
          0a6AH 0a6BH 0a6DH 0a6EH 0a8BH 0a8CH 0a9BH 0a9CH 0a9GH 0a19BH
          0a26AH 0a26BH 0a27AH 0a27BH 0a28AH 0a29BH 0a41H 0a1AH 0a1BH
- 1a5AH :
- 0a30CH :
```

D

Figure S2. (continued)



Haplotype list:

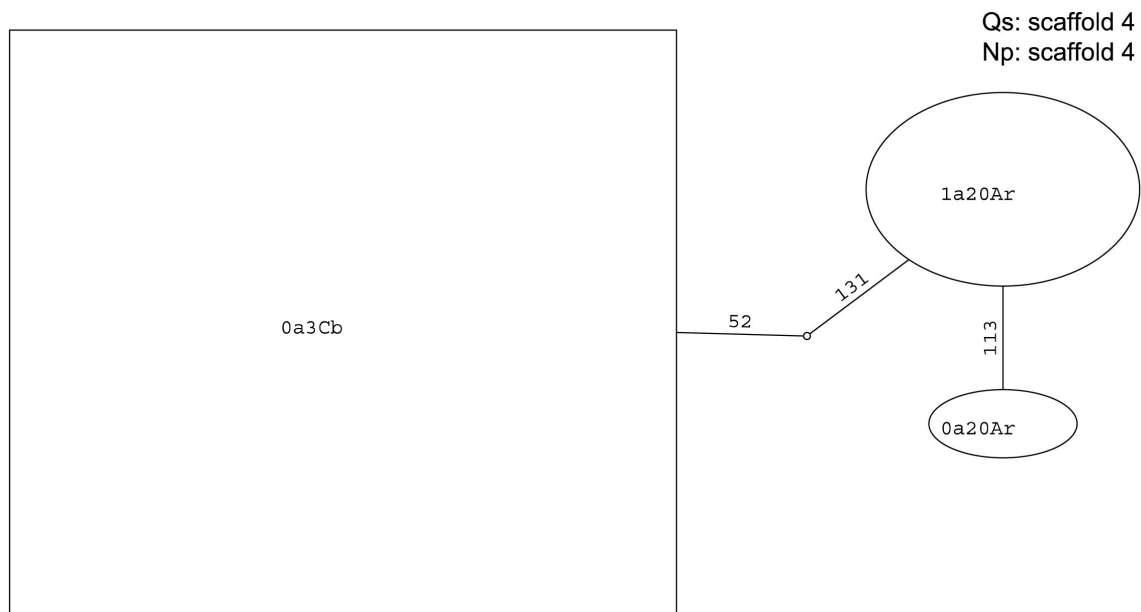
```

- 0a5Db : 1a5Db 0a6Cb 1a6Cb 0a6Fb 1a6Fb 0a7Db 1a7Db 0a8Db 1a8Db 0a9Ab
          1a9Ab 0a9Db 1a9Db 0a9Eb 1a9Eb 0a9Fb 1a9Fb 0a9Hb 1a9Hb 0a10Bb
          1a10Bb 0a11Bb 1a11Bb 0a13b 1a13b 0a18Ab 0a28Bb 1a28Bb 0a31Bb 1a31Bb
          0a31Cb 1a31Cb 0a31Db 1a31Db 1a32Bb 0a36b 1a36b 0a37Ab 1a37Ab 1a37Bb
          1a48b 0a49Ab 1a49Ab 1a50b 0a51b 1a51b 0a52Ab 1a52Ab 1a52Bb 0a2H
          0a4BH 0a5AH 0a5BH 0a5CH 0a5EH 0a6AH 0a6BH 0a6DH 0a6EH 0a7AH
          0a7CH 0a8AH 0a8BH 0a8CH 0a9BH 0a9CH 0a9GH 0a22H 0a27BH 0a28AH
          0a30AH 0a30CH 0a34H 0a1BH
- 0a15Ab : 1a15Ab 1a18Ab 0a18BH 0a19BH
- 0a32Bb : 0a35b 1a35b 0a37Bb 0a43b 1a43b 0a44b 1a44b 0a45Bb 1a45Bb 0a46b
          0a49Bb 1a49Bb 0a50b 0a29BH
- 0a39b : 1a39b 0a40b 1a40b 0a26AH 0a27AH 0a41H
- 1a46b :
- 0a48b :
- 0a52Bb : 0a53Ab
- 1a53Ab : 0a53Bb
- 1a53Bb :
- 0a20Ar : 1a20Ar 0a20Br 1a20Br 0a21Ar 1a21Ar 0a21Br 1a21Br 0a23Ar 1a23Ar 0a23Br
          1a23Br 0a24Ar 1a24Ar 0a25Ar 1a25Ar 0a25Br 1a25Br 0a30Br 1a30Br 0a30Dr
          1a30Dr 1a2H 1a4BH 1a5AH 1a5BH 1a5CH 1a5EH 1a6AH 1a6BH 1a6DH
          1a6EH 1a7AH 1a7CH 1a8AH 1a8BH 1a8CH 1a9BH 1a9CH 1a9GH 1a18BH
          1a19BH 1a22H 1a26AH 1a27AH 1a27BH 1a28AH 1a29BH 1a30AH 1a30CH 1a34H
          1a41H 1a1BH

```

F

Figure S2. (continued)



Haplotype list:

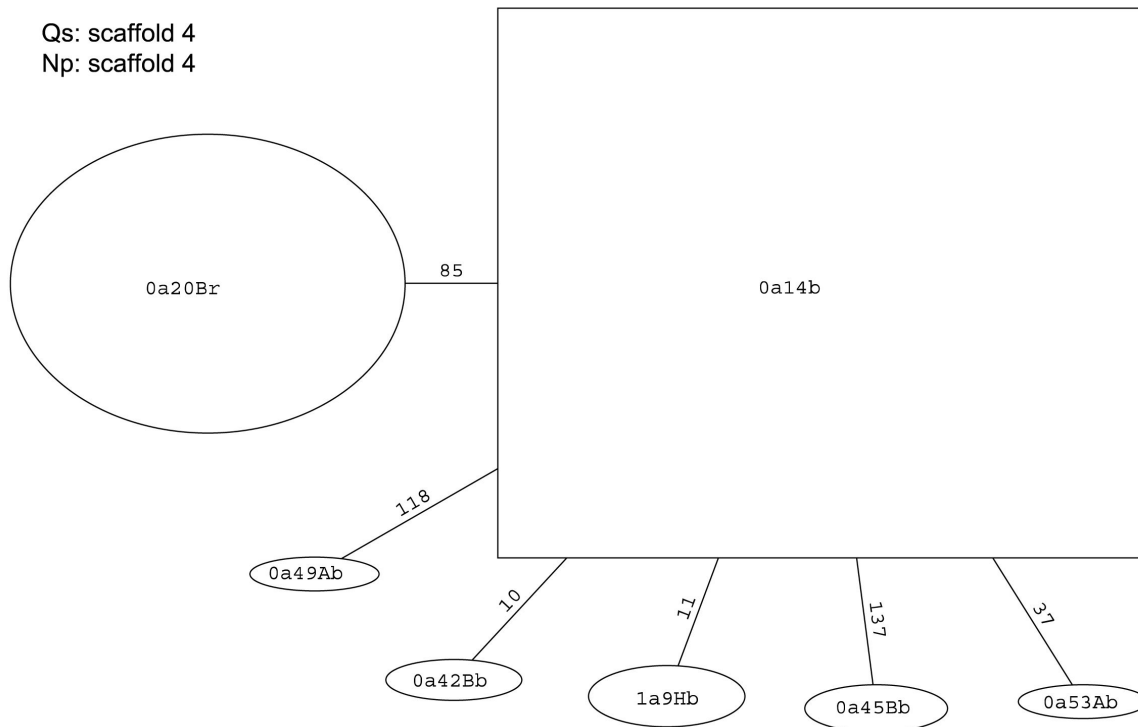
```

- 0a3Cb : 1a3Cb 0a5Db 1a5Db 0a6Cb 1a6Cb 0a6Fb 1a6Fb 0a8Db 1a8Db 0a9Db 1a9Db
           0a9Eb 1a9Eb 0a9Fb 1a9Fb 0a9Hb 1a9Hb 0a10Ab 1a10Ab 0a10Bb 1a10Bb 0a13b
           1a13b 0a15Ab 1a15Ab 0a15Bb 1a15Bb 0a18Ab 1a18Ab 0a28Bb 1a28Bb 0a31Ab 1a31Ab
           0a31Bb 1a31Bb 0a31Cb 1a31Cb 0a31Db 1a31Db 0a32Bb 1a32Bb 0a35b 1a35b 0a36b
           1a36b 0a37Ab 1a37Ab 0a37Bb 1a37Bb 0a40b 1a40b 0a42Bb 1a42Bb 0a45Ab 1a45Ab
           0a45Bb 1a45Bb 0a46b 1a46b 0a47b 1a47b 0a49Bb 1a49Bb 0a50b 1a50b 0a51b
           1a51b 0a52Ab 1a52Ab 0a52Bb 1a52Bb 0a53Ab 1a53Ab 0a42Ab 1a42Ab 0a3BH 0a5BH
           0a5CH 0a5EH 0a6AH 0a6BH 0a6EH 0a8AH 0a8BH 0a8CH 0a9BH 0a9CH 0a19AH
           0a19BH 0a22H 0a26BH 0a27AH 0a27BH 0a29BH 0a30AH 0a30CH 0a33H 0a34H 0a41H
- 0a20Ar : 0a21Ar 0a23Ar 1a23Ar 0a24Ar 0a25Ar 1a19AH 1a19BH 1a22H
- 1a20Ar : 0a20Br 1a20Br 1a21Ar 0a21Br 1a21Br 1a24Ar 1a25Ar 0a30Dr 1a30Dr 1a3BH 1a5BH
           1a5CH 1a5EH 1a6AH 1a6BH 1a6EH 1a8AH 1a8BH 1a8CH 1a9BH 1a9CH 1a26BH
           1a27AH 1a27BH 1a29BH 1a30AH 1a30CH 1a33H 1a34H 1a41H

```

G

Figure S2. (continued)



Haplotype list:

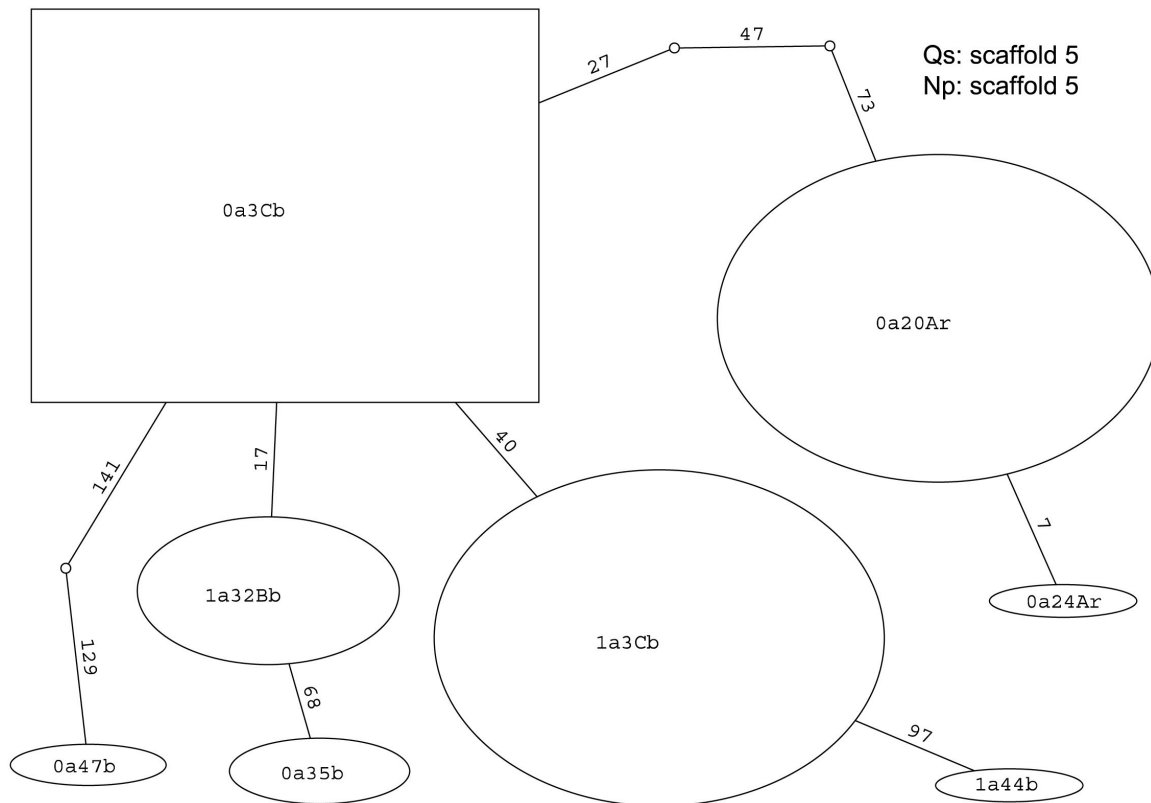
```

- 0a14b : 1a14b 0a3Cb 1a3Cb 0a5Db 1a5Db 0a6Cb 1a6Cb 0a6Fb 1a6Fb 0a8Db
          1a8Db 0a9Db 1a9Db 0a9Eb 1a9Eb 0a9Fb 1a9Fb 0a9Hb 0a10Ab 1a10Ab
          0a10Bb 0a13b 1a13b 0a15Ab 1a15Ab 0a15Bb 1a15Bb 0a31Ab 1a31Ab 0a31Bb
          1a31Bb 0a31Cb 1a31Cb 0a31Db 1a31Db 0a35b 1a35b 0a37Ab 1a37Ab 0a37Bb
          1a37Bb 0a39b 1a39b 0a40b 1a40b 0a46b 1a46b 1a47b 1a48b 0a49Bb
          1a49Bb 0a50b 1a50b 0a52Ab 1a52Ab 0a52Bb 1a52Bb 1a3BH 1a5AH 1a5BH
          1a6AH 1a6BH 1a6DH 1a6EH 1a8AH 1a8BH 1a8CH 1a9CH 1a9GH 1a19AH
          1a19BH 1a22H 1a27AH 1a27BH 1a29BH 1a32AH 1a41H
- 1a9Hb : 1a10Bb 1a5CH 1a9BH 1a30AH 1a30CH
- 0a42Bb : 1a42Bb 0a47b
- 0a45Bb : 1a45Bb 0a48b 1a33H
- 0a49Ab : 1a49Ab
- 0a53Ab : 1a53Ab
- 0a20Br : 1a20Br 0a21Br 1a21Br 0a23Ar 1a23Ar 0a23Br 1a23Br 0a24Ar 1a24Ar 0a25Ar
          1a25Ar 0a25Br 1a25Br 0a30Dr 1a30Dr 0a3BH 0a5AH 0a5BH 0a5CH 0a6AH
          0a6BH 0a6DH 0a6EH 0a8AH 0a8BH 0a8CH 0a9BH 0a9CH 0a9GH 0a19AH
          0a19BH 0a22H 0a27AH 0a27BH 0a29BH 0a30AH 0a30CH 0a32AH 0a33H 0a41H

```

H

Figure S2. (continued)



Haplotype list:

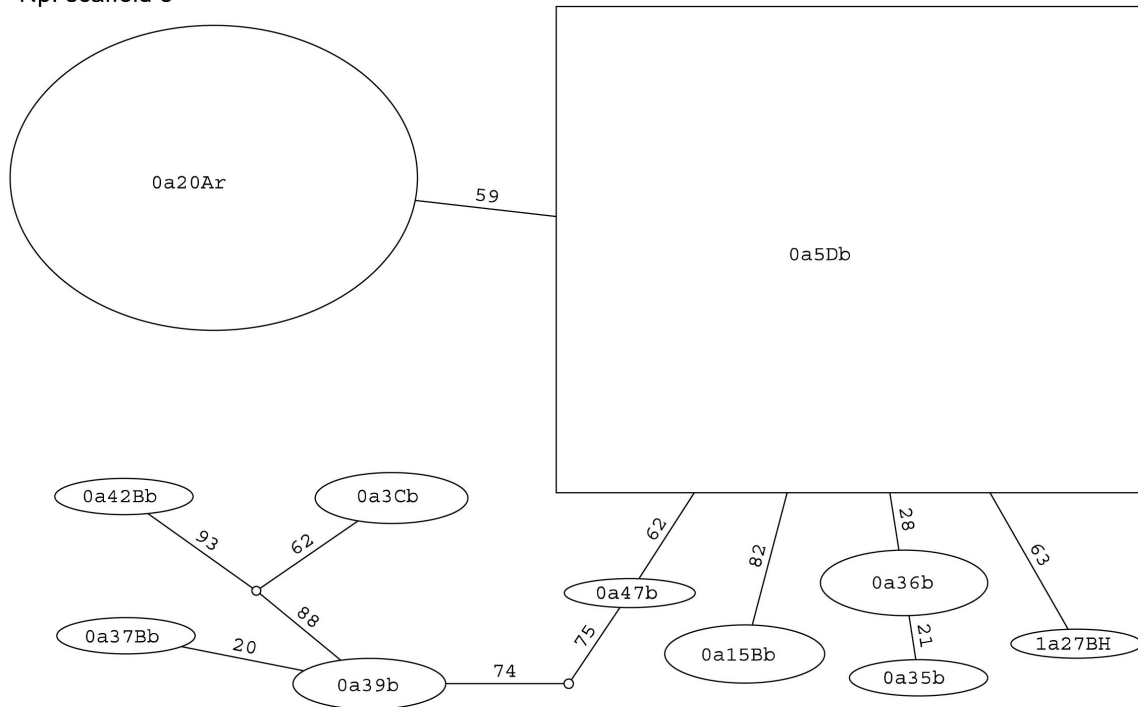
- 0a3Cb : 0a5Db 1a5Db 0a6Cb 1a6Cb 0a8Db 1a8Db 0a9Ab 1a9Ab 0a9Db 1a9Db 0a9Eb 1a9Eb 0a9Fb 1a9Fb 0a9Hb 1a9Hb 0a10Ab 1a10Ab 0a10Bb 1a10Bb 0a13b 1a13b 0a28Bb 0a43b 1a43b 0a44b 0a45Bb 1a45Bb 0a48b 1a48b 1a4AH 1a5AH 1a5BH 1a5CH 1a5EH 1a6AH 1a6DH 1a8BH 1a8CH 1a9CH 1a9GH 1a28AH 1a30AH 1a30CH
- 1a3Cb : 0a15Ab 1a15Ab 0a15Bb 1a15Bb 0a18Ab 1a18Ab 1a28Bb 0a31Ab 1a31Ab 0a31Bb 1a31Bb 0a31Cb 1a31Cb 0a31Db 1a31Db 0a32Bb 0a36b 0a37Ab 0a37Bb 0a39b 0a40b 0a42Bb 1a42Bb 0a49Bb 0a50b 0a51b 1a51b 0a53Ab 1a3BH 1a3AH 1a19AH 1a19BH 1a27AH 1a27BH 1a29BH 1a33H 1a1BH
- 1a32Bb : 1a37Ab 1a37Bb 0a49Ab 1a49Ab 1a49Bb 1a50b 0a52Ab 1a52Ab 0a52Bb 1a52Bb 1a53Ab 0a53Bb 1a53Bb 1a32AH
- 0a35b : 1a35b 1a36b 1a39b 1a40b
- 1a44b :
- 0a47b : 1a47b
- 0a20Ar : 1a20Ar 0a21Ar 1a21Ar 0a23Ar 1a23Ar 0a23Br 1a23Br 1a24Ar 0a30Br 1a30Br 0a30Dr 1a30Dr 0a3BH 0a4AH 0a5AH 0a5BH 0a5CH 0a5EH 0a6AH 0a6DH 0a8BH 0a8CH 0a9CH 0a9GH 0a3AH 0a19AH 0a19BH 0a27AH 0a27BH 0a28AH 0a29BH 0a30AH 0a30CH 0a32AH 0a33H 0a1BH
- 0a24Ar :

I

Figure S2. (continued)

Qs: scaffold 5

Np: scaffold 5



Haplotype list:

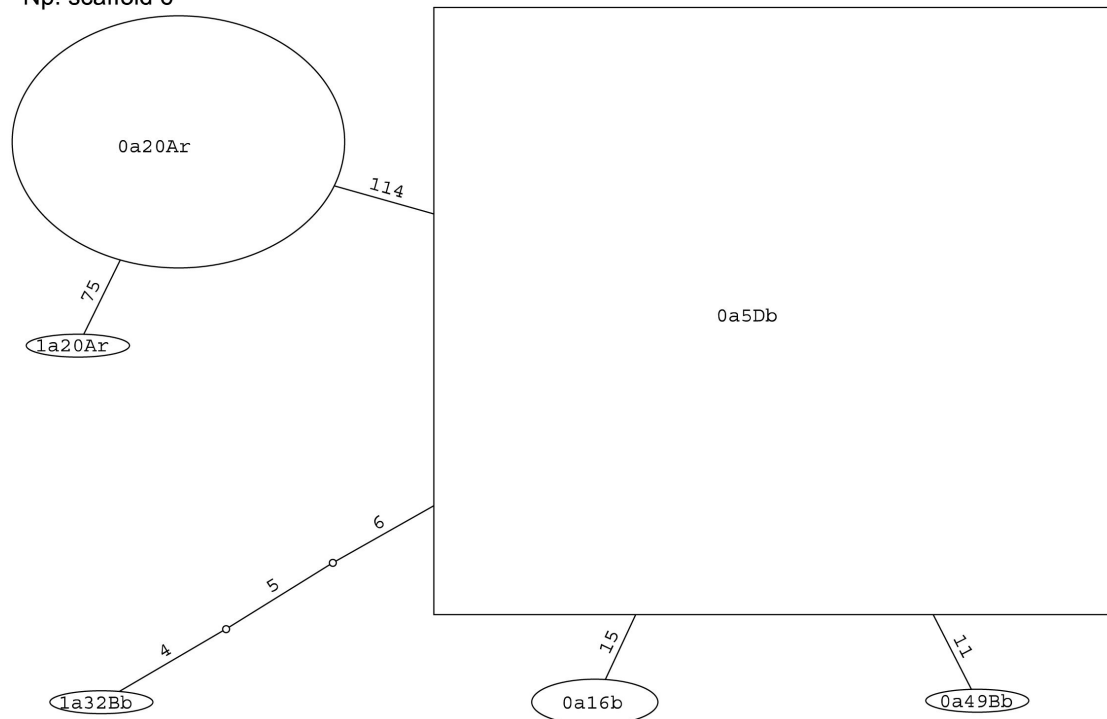
- 0a3Cb : 1a3Cb 1a3BH 1a3AH
- 0a5Db : 1a5Db 0a6Cb 1a6Cb 0a8Db 1a8Db 0a9Ab 1a9Ab 0a9Db 1a9Db 0a9Eb 1a9Eb
0a9Fb 1a9Fb 0a9Hb 1a9Hb 0a13b 1a13b 0a17b 1a17b 0a28Bb 1a28Bb 0a31Ab
1a31Ab 0a31Bb 1a31Bb 0a31Cb 1a31Cb 0a31Db 1a31Db 0a32Bb 1a32Bb 1a36b 0a40b
1a40b 0a44b 1a44b 0a45Bb 1a45Bb 1a47b 0a48b 1a48b 1a51b 0a52Bb 1a52Bb
0a53Ab 1a53Ab 0a53Bb 1a53Bb 1a5AH 1a5BH 1a5CH 1a5EH 1a6BH 1a6EH 1a8AH
1a8BH 1a8CH 1a9BH 1a9CH 1a9GH 1a27AH 1a30AH 1a30CH
- 0a15Bb : 1a15Bb 1a19AH 1a19BH 1a22H
- 0a35b : 1a35b
- 0a36b : 0a50b 1a50b 0a51b 0a29BH 0a32AH
- 0a37Bb : 1a37Bb
- 0a39b : 1a39b 0a45Ab 1a45Ab
- 0a42Bb : 1a42Bb
- 0a47b :
- 0a20Ar : 1a20Ar 0a20Br 1a20Br 0a21Br 1a21Br 0a23Ar 1a23Ar 0a23Br 1a23Br 0a24Ar 1a24Ar
0a25Br 1a25Br 0a30Dr 1a30Dr 0a3BH 0a5AH 0a5BH 0a5CH 0a5EH 0a6BH 0a6EH
0a8AH 0a8BH 0a8CH 0a9BH 0a9CH 0a9GH 0a3AH 0a19AH 0a19BH 0a22H 0a27AH
0a27BH 1a29BH 0a30AH 0a30CH 1a32AH
- 1a27BH :

J

Figure S2. (continued)

Qs: scaffold 6

Np: scaffold 6

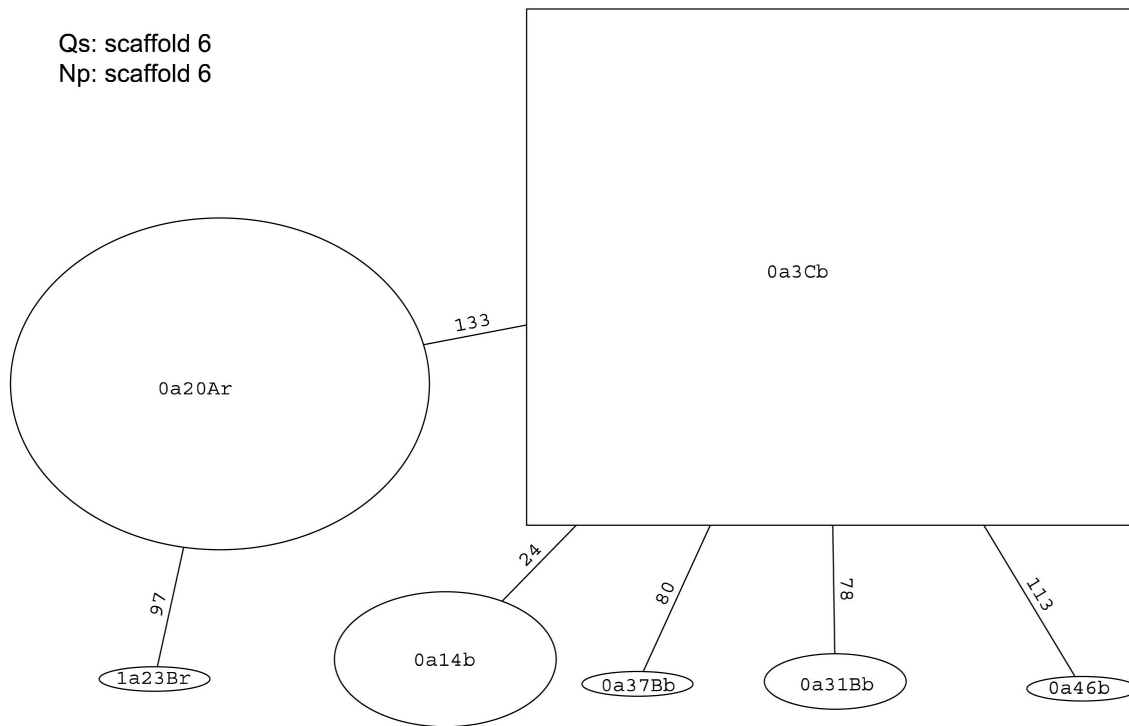


Haplotype list:

```
- 0a5Db : 1a5Db 0a6Cb 1a6Cb 0a6Fb 1a6Fb 0a7Bb 1a7Bb 0a8Db 1a8Db 0a9Ab 1a9Ab
          0a9Db 1a9Db 0a9Eb 1a9Eb 0a9Fb 1a9Fb 0a9Hb 1a9Hb 0a11Ab 1a11Ab 0a11Bb
          1a11Bb 0a17b 0a28Bb 1a28Bb 0a31Ab 1a31Ab 0a31Bb 1a31Bb 0a31Cb 1a31Cb 0a31Db
          1a31Db 0a32Bb 0a35b 1a35b 0a36b 1a36b 0a37Ab 1a37Ab 0a37Bb 1a37Bb 0a38Ab
          1a38Ab 0a38Bb 1a38Bb 0a39b 1a39b 0a40b 1a40b 0a42Bb 1a42Bb 0a43b 1a43b
          0a44b 1a44b 0a45Ab 1a45Ab 0a45Bb 1a45Bb 0a46b 1a46b 0a47b 1a47b 0a48b
          1a48b 0a49Ab 1a49Ab 1a49Bb 0a53Ab 1a53Ab 0a53Bb 1a53Bb 0a42Ab 1a42Ab 0a2H
          0a4BH 0a5AH 0a5BH 0a5EH 0a6AH 0a6BH 0a6DH 0a6EH 0a7CH 0a8BH 0a8CH
          0a9BH 0a9CH 0a9GH 0a18BH 0a26AH 0a26BH 0a27AH 0a27BH 0a29AH 0a29BH 0a30AH
          0a30CH 0a41H 0a1AH
- 0a16b : 1a16b 1a17b 0a18Ab 1a18Ab
- 1a32Bb :
- 0a49Bb :
- 0a20Ar : 0a21Br 1a21Br 0a23Ar 1a23Ar 0a23Br 1a23Br 0a24Ar 1a24Ar 0a25Ar 1a25Ar 0a30Br
          1a30Br 0a30Dr 1a30Dr 1a2H 1a4BH 1a5AH 1a5BH 1a5EH 1a6AH 1a6BH 1a6DH
          1a6EH 1a7CH 1a8BH 1a8CH 1a9BH 1a9CH 1a9GH 1a18BH 1a26AH 1a26BH 1a27AH
          1a27BH 1a29AH 1a29BH 1a30AH 1a30CH 1a41H 1a1AH
- 1a20Ar :
```

K

Figure S2. (continued)

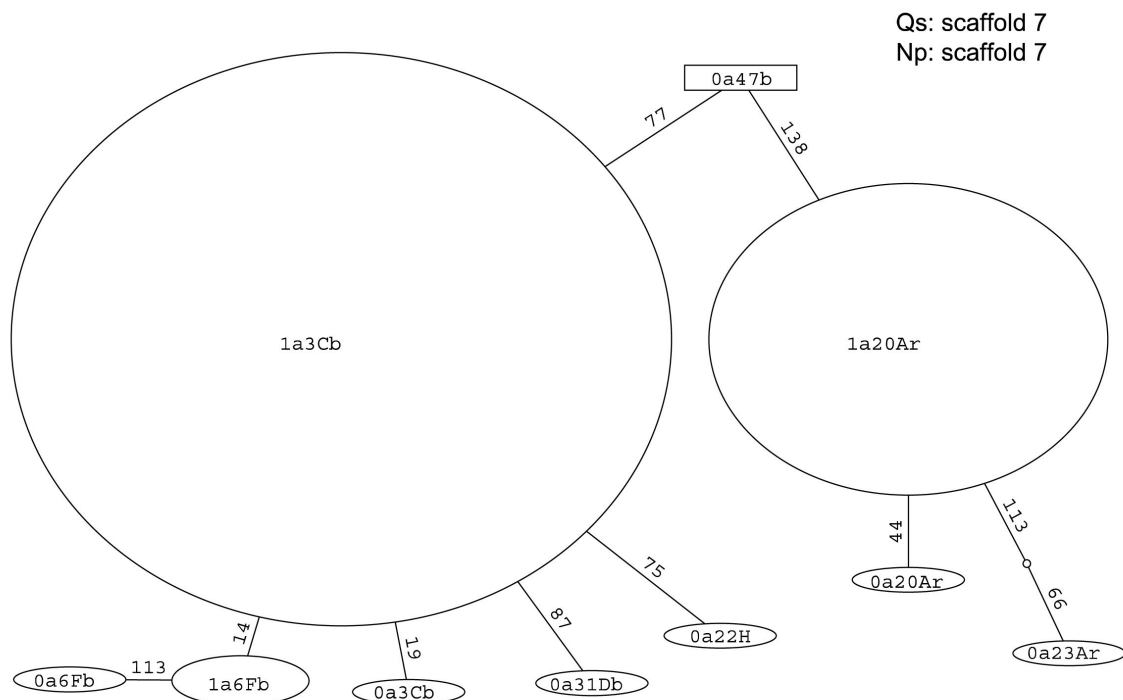


Haplotype list:

```
- 0a14b : 1a14b 0a15Ab 1a15Ab 0a15Bb 1a15Bb 1a37Ab 1a37Bb 1a45Bb 0a48b 1a48b 0a49Bb
          1a49Bb 0a53Ab 1a53Ab 0a53Bb 1a53Bb 1a19BH 1a22H
- 0a3Cb : 1a3Cb 0a5Db 1a5Db 0a6Cb 1a6Cb 0a6Fb 1a6Fb 0a8Db 1a8Db 0a9Ab 1a9Ab
          0a9Db 1a9Db 0a9Eb 1a9Eb 0a9Fb 1a9Fb 0a9Hb 1a9Hb 0a10Ab 1a10Ab 0a10Bb
          1a10Bb 0a12b 1a12b 0a13b 1a13b 0a32Bb 1a32Bb 0a35b 1a35b 0a36b 1a36b
          0a37Ab 0a39b 1a39b 0a40b 1a40b 0a42Bb 1a42Bb 0a45Ab 1a45Ab 0a45Bb 1a46b
          0a50b 1a50b 0a52Ab 1a52Ab 0a52Bb 1a52Bb 0a42Ab 1a42Ab 1a3BH 1a4AH 1a5AH
          1a5BH 1a5CH 1a5EH 1a6AH 1a6BH 1a6DH 1a6EH 1a8AH 1a8BH 1a8CH 1a9BH
          1a9CH 1a9GH 1a3AH 1a19AH 1a26BH 1a27AH 1a27BH 1a28AH 1a29BH 1a30AH 1a30CH
          1a32AH 1a33H 1a34H
- 0a31Bb : 1a31Bb 0a31Cb 1a31Cb 0a31Db 1a31Db
- 0a37Bb :
- 0a46b :
- 0a20Ar : 1a20Ar 0a20Br 1a20Br 0a21Ar 1a21Ar 0a21Br 1a21Br 0a23Ar 1a23Ar 0a23Br 0a24Ar
          1a24Ar 0a24Br 1a24Br 0a25Ar 1a25Ar 0a25Br 1a25Br 0a30Dr 1a30Dr 0a3BH 0a4AH
          0a5AH 0a5BH 0a5CH 0a5EH 0a6AH 0a6BH 0a6DH 0a6EH 0a8AH 0a8BH 0a8CH
          0a9BH 0a9CH 0a9GH 0a3AH 0a19AH 0a19BH 0a22H 0a26BH 0a27AH 0a27BH 0a28AH
          0a29BH 0a30AH 0a30CH 0a32AH 0a33H 0a34H
- 1a23Br :
```

L

Figure S2. (continued)

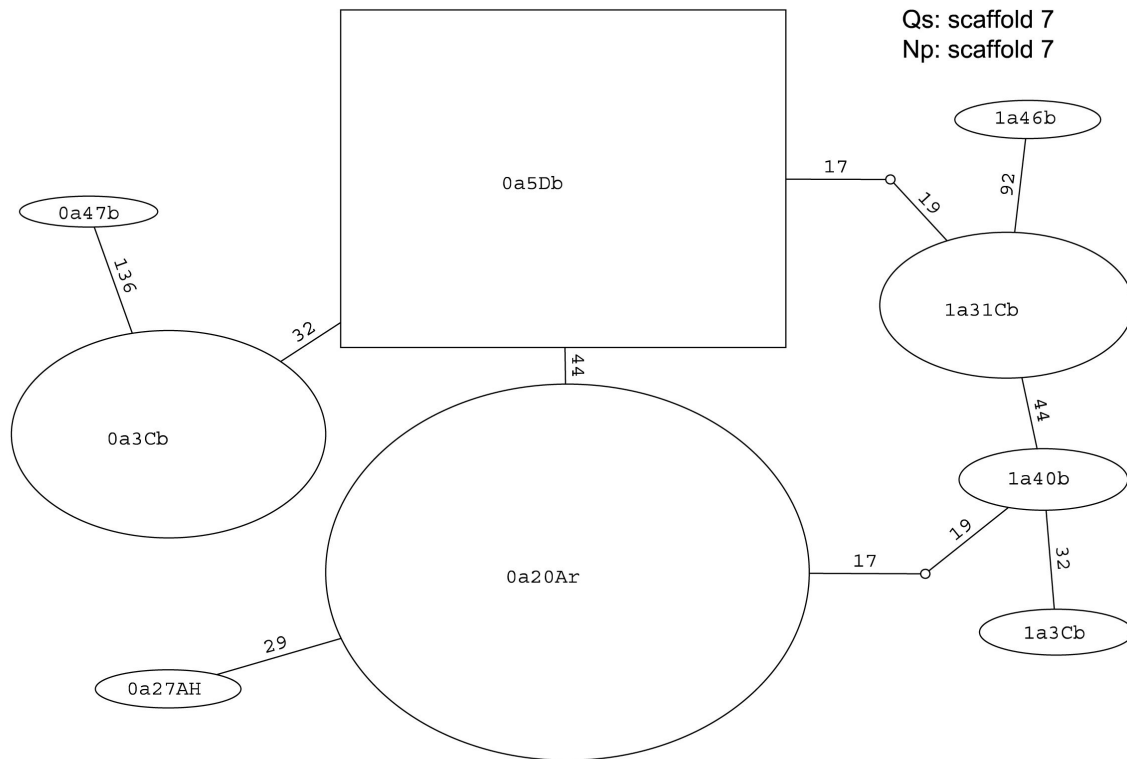


Haplotype list:

```
- 0a3Cb :
- 1a3Cb : 0a8Db 1a8Db 0a9Db 1a9Db 0a9Eb 1a9Eb 0a9Fb 1a9Fb 0a9Hb 1a9Hb 1a10Ab
          0a13b 1a13b 0a15Ab 1a15Ab 0a15Bb 1a15Bb 0a18Ab 1a18Ab 0a31Bb 1a31Bb 0a31Cb
          1a31Cb 1a31Db 0a35b 1a35b 0a36b 1a36b 0a37Ab 1a37Ab 0a37Bb 1a37Bb 0a38Ab
          1a38Ab 0a39b 1a39b 0a40b 1a40b 0a45Ab 1a45Ab 0a45Bb 1a45Bb 0a46b 1a46b
          1a47b 0a48b 1a48b 0a49Ab 1a49Ab 0a49Bb 1a49Bb 0a50b 1a50b 0a51b 1a51b
          0a52Ab 1a52Ab 0a52Bb 1a52Bb 0a53Ab 1a53Ab 0a53Bb 1a53Bb 1a3BH 1a4AH 1a5AH
          1a5EH 1a6AH 1a6BH 1a6EH 1a8BH 1a8CH 1a9BH 1a9CH 1a9GH 1a19AH 1a19BH
          1a26BH 1a27AH 1a27BH 1a29AH 1a29BH 1a30AH 1a30CH 1a32AH 1a33H 1a34H 1a41H
- 0a6Fb :
- 1a6Fb : 0a10Ab 0a5BH 0a5CH 0a8AH
- 0a31Db :
- 0a47b :
- 0a20Ar :
- 1a20Ar : 0a20Br 1a20Br 0a21Ar 1a21Ar 0a21Br 1a21Br 1a23Ar 0a24Ar 1a24Ar 0a24Br 1a24Br
          0a25Ar 1a25Ar 0a25Br 1a25Br 0a30Dr 1a30Dr 0a3BH 0a4AH 0a5AH 1a5BH 1a5CH
          0a5EH 0a6AH 0a6BH 0a6EH 1a8AH 0a8BH 0a8CH 0a9BH 0a9CH 0a9GH 0a19AH
          0a19BH 1a22H 0a26BH 0a27AH 0a27BH 0a29AH 0a29BH 0a30AH 0a30CH 0a32AH 0a33H
          0a34H 0a41H
- 0a23Ar :
- 0a22H :
```

M

Figure S2. (continued)



Haplotype list:

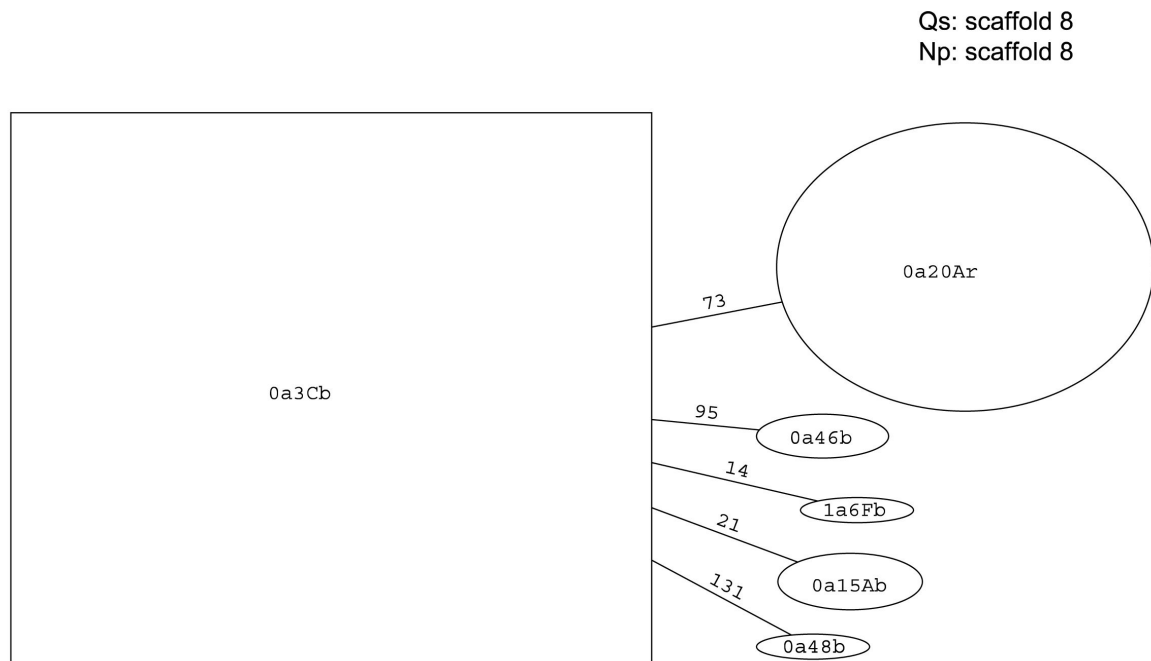
```

- 0a3Cb : 0a12b 1a12b 0a15Ab 1a15Ab 0a15Bb 1a15Bb 1a36b 1a37Ab 0a37Bb 0a45Ab 0a45Bb
          0a46b 0a49Bb 1a49Bb 0a53Bb 1a53Bb 0a42Ab 1a3BH 1a3AH 1a19AH 1a19BH 1a22H
          1a26BH
- 1a3Cb : 0a13b 1a13b
- 0a5Db : 1a5Db 0a6Cb 1a6Cb 0a8Db 1a8Db 0a9Ab 1a9Ab 0a9Db 1a9Db 0a9Fb 1a9Fb
          0a9Hb 1a9Hb 0a10Ab 1a10Ab 0a10Bb 1a10Bb 0a31Ab 1a31Ab 0a31Bb 1a31Bb 0a31Cb
          0a31Db 1a31Db 0a36b 0a37Ab 0a4AH 0a5BH 0a5CH 0a5EH 0a6AH 0a6BH 0a6DH
          0a8AH 0a8BH 0a8CH 0a9CH 0a9GH 0a29BH 0a30CH
- 1a31Cb : 0a35b 1a35b 1a37Bb 0a38Ab 1a38Ab 0a39b 1a39b 0a40b 0a48b 0a51b 1a51b
          1a27AH 1a27BH 1a33H 1a34H
- 1a40b : 1a45Ab 1a45Bb 1a48b 1a42Ab
- 1a46b : 1a47b
- 0a47b :
- 0a20Ar : 1a20Ar 0a20Br 1a20Br 0a21Ar 1a21Ar 0a21Br 1a21Br 0a23Ar 1a23Ar 0a23Br 1a23Br
          0a24Ar 1a24Ar 0a24Br 1a24Br 0a25Ar 1a25Ar 0a25Br 1a25Br 0a30Br 1a30Br 0a30Dr
          1a30Dr 0a3BH 1a4AH 1a5BH 1a5CH 1a5EH 1a6AH 1a6BH 1a6DH 1a8AH 1a8BH
          1a8CH 1a9CH 1a9GH 0a3AH 0a19AH 0a19BH 0a22H 0a26BH 1a29BH 1a30CH 0a33H
          0a34H
- 0a27AH : 0a27BH

```

N

Figure S2. (continued)



Haplotype list:

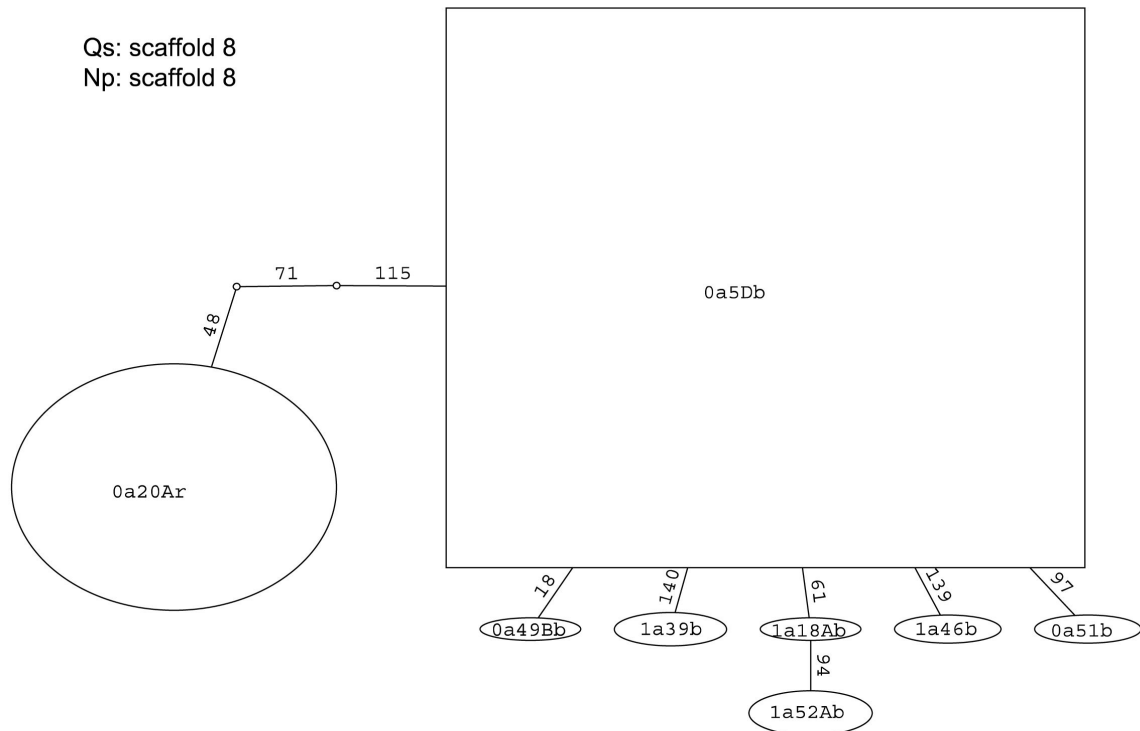
```

- 0a3Cb : 1a3Cb 0a5Db 1a5Db 0a6Fb 0a8Db 1a8Db 0a9Ab 1a9Ab 0a9Db 1a9Db
          0a9Eb 1a9Eb 0a9Fb 1a9Fb 0a9Hb 1a9Hb 0a10Ab 1a10Ab 0a10Bb 1a10Bb
          0a13b 1a13b 0a31Ab 1a31Ab 0a31Bb 1a31Bb 0a31Cb 1a31Cb 0a31Db 1a31Db
          0a32Bb 1a32Bb 0a36b 1a36b 0a37Ab 1a37Ab 0a37Bb 1a37Bb 0a39b 1a39b
          0a40b 1a40b 0a44b 1a44b 0a45Ab 1a45Ab 0a45Bb 1a45Bb 1a48b 0a49Ab
          1a49Ab 0a50b 1a50b 0a51b 1a51b 0a52Ab 1a52Ab 0a53Ab 1a53Ab 0a53Bb
          1a53Bb 0a3BH 0a5AH 0a5BH 0a5CH 0a5EH 0a6BH 0a6DH 0a6EH 0a8AH
          0a8BH 0a8CH 0a9BH 0a9CH 0a9GH 0a22H 0a27AH 0a27BH 0a29BH 0a30AH
          0a30CH 0a33H 0a34H 0a41H
- 1a6Fb :
- 0a15Ab : 1a15Ab 0a15Bb 1a15Bb 1a19AH 1a19BH
- 0a46b : 1a46b 0a47b 1a47b
- 0a48b :
- 0a20Ar : 1a20Ar 0a20Br 1a20Br 0a21Br 1a21Br 0a23Ar 1a23Ar 0a23Br 1a23Br 0a24Ar
          1a24Ar 0a25Ar 1a25Ar 0a30Br 1a30Br 0a30Dr 1a30Dr 1a3BH 1a5AH 1a5BH
          1a5CH 1a5EH 1a6BH 1a6DH 1a6EH 1a8AH 1a8BH 1a8CH 1a9BH 1a9CH
          1a9GH 0a19AH 0a19BH 1a22H 1a27AH 1a27BH 1a29BH 1a30AH 1a30CH 1a33H
          1a34H 1a41H

```

O

Figure S2. (continued)



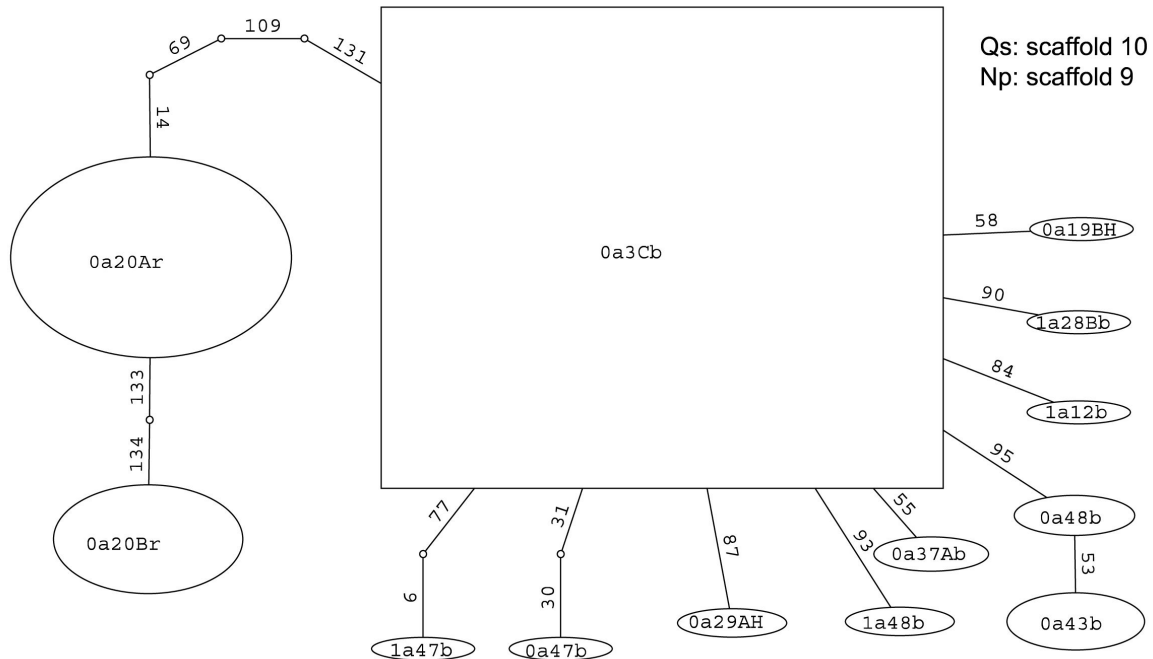
Haplotype list:

```
- 0a5Db : 1a5Db 0a6Cb 1a6Cb 0a6Fb 1a6Fb 0a7Bb 1a7Bb 0a7Db 1a7Db 0a8Db 1a8Db
          0a9Ab 1a9Ab 0a9Db 1a9Db 0a9Eb 1a9Eb 0a9Fb 1a9Fb 0a9Hb 1a9Hb 0a10Bb
          1a10Bb 0a11Bb 1a11Bb 0a13b 1a13b 0a15Ab 1a15Ab 0a17b 1a17b 0a18Ab 0a28Bb
          1a28Bb 0a31Ab 1a31Ab 0a31Bb 1a31Bb 0a31Cb 1a31Cb 0a31Db 1a31Db 0a32Bb 1a32Bb
          0a35b 1a35b 0a37Ab 1a37Ab 0a37Bb 1a37Bb 0a38Bb 1a38Bb 0a39b 0a40b 1a40b
          0a42Bb 0a43b 1a43b 0a44b 0a45Bb 1a45Bb 0a46b 0a48b 1a48b 0a49Ab 1a49Ab
          1a49Bb 0a50b 1a50b 0a52Ab 0a42Ab 1a2H 1a5AH 1a5BH 1a5EH 1a6AH 1a6BH
          1a6DH 1a6EH 1a8AH 1a8BH 1a8CH 1a9BH 1a9GH 1a19AH 1a19BH 1a22H 1a27AH
          1a27BH 1a29BH 1a30AH 1a30CH 1a33H 1a34H 1a41H 1a1BH

- 1a18Ab :
- 1a39b : 1a42Bb 1a44b
- 1a46b : 1a42Ab
- 0a49Bb :
- 0a51b : 1a51b
- 1a52Ab : 0a53Ab 1a53Ab 0a53Bb 1a53Bb
- 0a20Ar : 1a20Ar 0a21Br 1a21Br 0a23Ar 1a23Ar 0a23Br 1a23Br 0a24Ar 1a24Ar 0a24Br 1a24Br
          0a25Br 1a25Br 0a30Dr 1a30Dr 0a2H 0a5AH 0a5BH 0a5EH 0a6AH 0a6BH 0a6DH
          0a6EH 0a8AH 0a8BH 0a8CH 0a9BH 0a9GH 0a19AH 0a19BH 0a22H 0a27AH 0a27BH
          0a29BH 0a30AH 0a30CH 0a33H 0a34H 0a41H 0a1BH
```

P

Figure S2. (continued)

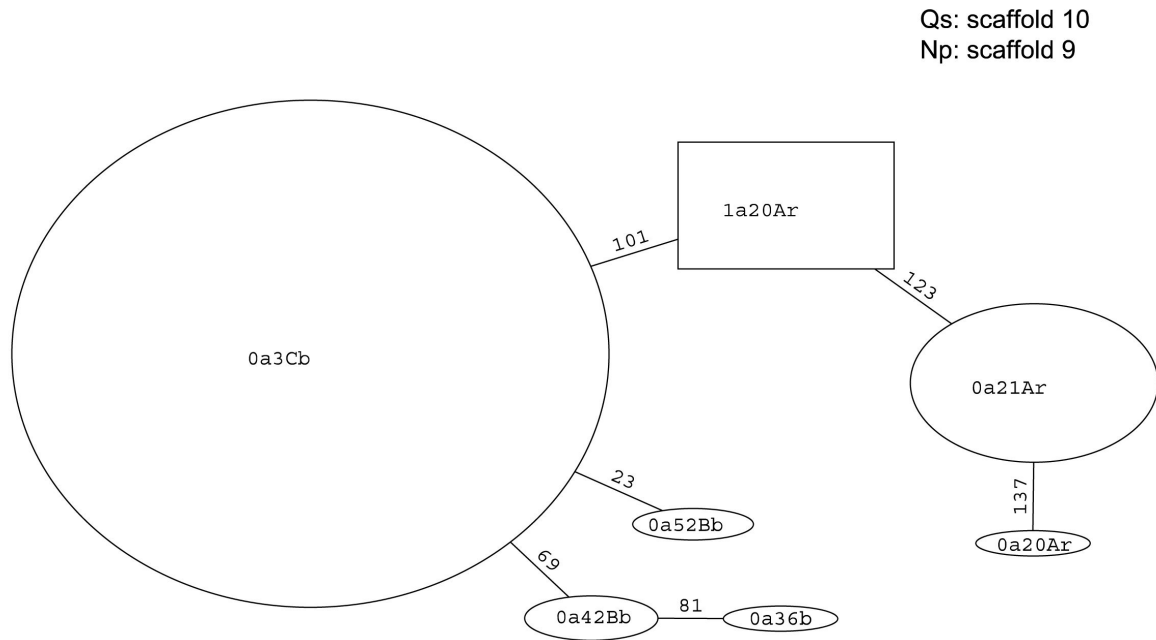


Haplotype list:

- 0a3Cb : 1a3Cb 0a5Db 1a5Db 0a6Cb 1a6Cb 0a6Fb 1a6Fb 0a8Db 1a8Db 0a9Ab 1a9Ab
0a9Db 1a9Db 0a9Eb 1a9Eb 0a9Fb 1a9Fb 0a9Hb 1a9Hb 0a10Ab 1a10Ab 0a12b
0a15Ab 1a15Ab 0a15Bb 1a15Bb 0a28Bb 0a31Ab 1a31Ab 0a31Bb 1a31Bb 0a31Cb 1a31Cb
0a31Db 1a31Db 0a36b 1a36b 1a37Ab 0a38Ab 1a38Ab 0a38Bb 1a38Bb 0a39b 1a39b
0a40b 1a40b 0a42Bb 1a42Bb 0a46b 1a46b 0a50b 1a50b 0a51b 1a51b 0a52Ab
1a52Ab 0a52Bb 1a52Bb 0a53Ab 1a53Ab 0a3BH 0a5AH 0a5BH 0a5CH 0a5EH 0a6AH
0a6BH 0a6DH 0a6EH 0a8AH 0a8BH 0a8CH 0a9GH 0a19AH 0a22H 0a26AH 0a30CH
0a33H 0a34H 0a41H
- 1a12b :
- 1a28Bb :
- 0a37Ab : 0a45Bb 0a49Ab
- 0a43b : 1a43b 0a45Ab 1a45Ab 1a45Bb 0a42Ab 1a42Ab
- 0a47b :
- 1a47b :
- 0a48b : 0a26BH 0a27AH 0a27BH
- 1a48b : 1a49Ab
- 0a20Ar : 1a20Ar 1a20Br 0a21Ar 1a21Ar 0a21Br 1a21Br 0a23Ar 1a23Ar 0a23Br 1a23Br 0a24Ar
1a24Ar 0a24Br 1a24Br 0a25Br 1a25Br 0a30Br 1a30Br 0a30Dr 1a30Dr 1a3BH 1a26AH
1a26BH 1a27AH 1a27BH 1a29AH 1a29BH 1a30CH 1a33H 1a34H 1a41H
- 0a20Br : 1a5AH 1a5BH 1a5CH 1a5EH 1a6AH 1a6BH 1a6DH 1a6EH 1a8AH 1a8BH 1a8CH
1a9GH 1a19AH 1a19BH 1a22H
- 0a19BH :
- 0a29AH : 0a29BH

Q

Figure S2. (continued)



Haplotype list:

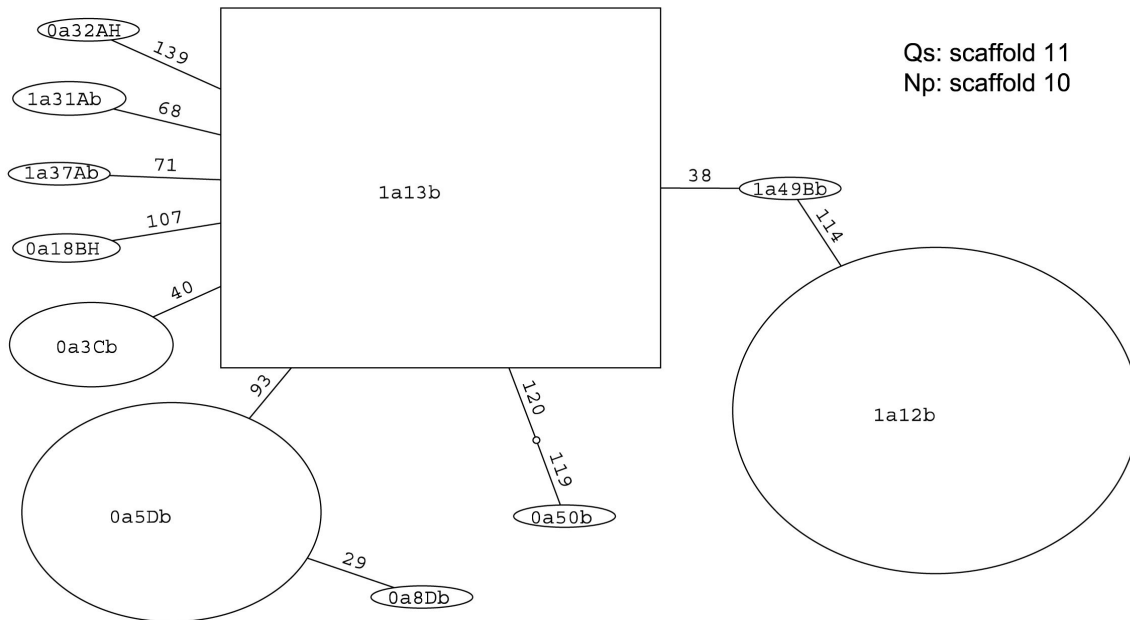
```

- 0a3Cb : 1a3Cb 0a5Db 1a5Db 0a6Cb 1a6Cb 0a8Db 1a8Db 0a9Ab 1a9Ab 0a9Db
          1a9Db 0a9Eb 1a9Eb 0a9Hb 1a9Hb 0a10Ab 1a10Ab 0a10Bb 1a10Bb 0a12b
          1a12b 0a13b 1a13b 0a15Ab 1a15Ab 0a15Bb 1a15Bb 0a31Ab 1a31Ab 0a31Bb
          1a31Bb 0a31Cb 1a31Cb 0a31Db 1a31Db 0a32Bb 1a32Bb 1a36b 0a40b 1a40b
          1a42Bb 0a45Ab 1a45Ab 0a46b 1a46b 0a47b 1a47b 0a48b 1a48b 0a49Ab
          1a49Ab 0a50b 1a50b 0a51b 1a51b 0a4AH 0a5AH 0a5CH 0a5EH 0a6AH
          0a6BH 0a6EH 0a8AH 0a8BH 0a8CH 0a9BH 0a9CH 0a9GH 0a3AH 0a22H
          0a29AH 0a29BH 0a30AH 0a30CH 0a33H 0a34H
- 0a36b :
- 0a42Bb : 0a27AH 0a27BH 0a32AH
- 0a52Bb : 1a52Bb
- 0a20Ar :
- 1a20Ar : 0a20Br 1a20Br 1a4AH 1a5AH 1a5CH 1a5EH 1a6AH 1a6BH 1a6EH 1a8AH
          1a8BH 1a8CH 1a9BH 1a9CH 1a9GH 1a22H
- 0a21Ar : 1a21Ar 0a23Ar 1a23Ar 0a23Br 1a23Br 0a24Ar 1a24Ar 0a30Br 1a30Br 0a30Dr
          1a30Dr 1a3AH 1a27AH 1a27BH 1a29AH 1a29BH 1a30AH 1a30CH 1a32AH 1a33H
          1a34H

```

R

Figure S2. (continued)



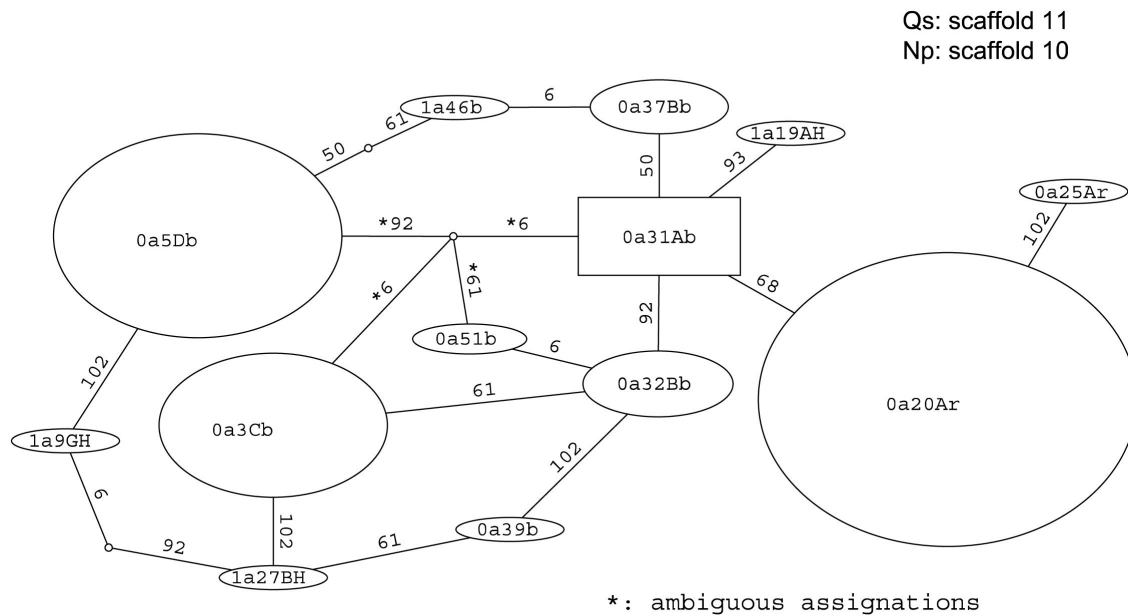
Haplotype list:

```

- 0a3Cb : 1a3Cb 0a12b 0a13b 0a31Ab 0a31Bb 0a31Cb 1a31Cb 0a31Db 0a2H 0a3BH 0a3AH
- 0a5Db : 1a5Db 0a6Cb 1a6Cb 0a6Fb 1a6Fb 0a7Db 1a7Db 1a8Db 0a9Ab 1a9Ab 0a9Eb
          1a9Eb 0a9Fb 1a9Fb 0a9Hb 1a9Hb 0a10Bb 1a10Bb 1a15Ab 0a5AH 0a5BH 0a5CH
          0a5EH 0a6AH 0a6BH 0a6EH 0a8AH 0a8BH 0a8CH 0a9BH 0a9CH 0a9GH 0a28AH
          0a30AH 0a30CH
- 0a8Db :
- 1a12b : 0a20Br 1a20Br 0a21Ar 1a21Ar 0a21Br 1a21Br 0a23Ar 1a23Ar 0a23Br 1a23Br 0a24Ar
          1a24Ar 0a24Br 1a24Br 0a25Ar 1a25Ar 0a25Br 1a25Br 0a30Br 1a30Br 0a30Dr 1a30Dr
          1a2H 1a3BH 1a5AH 1a5BH 1a5CH 1a5EH 1a6AH 1a6BH 1a6EH 1a8AH 1a8BH
          1a8CH 1a9BH 1a9CH 1a9GH 1a3AH 1a18BH 1a19BH 1a22H 1a26AH 1a26BH 1a27AH
          1a27BH 1a28AH 1a29AH 1a29BH 1a30AH 1a30CH 1a32AH 1a33H 1a34H 1a41H
- 1a13b : 0a15Ab 0a15Bb 1a15Bb 0a18Ab 1a18Ab 0a32Bb 1a32Bb 0a35b 1a35b 0a36b 1a36b
          0a37Ab 0a37Bb 1a37Bb 0a38Ab 1a38Ab 0a38Bb 1a38Bb 0a39b 1a39b 0a40b 1a40b
          0a42Bb 1a42Bb 0a43b 1a43b 0a45Ab 1a45Ab 0a45Bb 1a45Bb 0a47b 1a47b 0a48b
          1a48b 0a49Ab 1a49Ab 0a49Bb 1a50b 0a51b 1a51b 0a52Ab 1a52Ab 0a52Bb 1a52Bb
          0a53Ab 1a53Ab 0a53Bb 1a53Bb 0a42Ab 1a42Ab 0a22H 0a26AH 0a26BH 0a27AH 0a27BH
          0a29AH 0a29BH 0a33H 0a34H 0a41H
- 1a31Ab : 1a31Bb 1a31Db
- 1a37Ab :
- 1a49Bb :
- 0a50b :
- 0a18BH : 0a19BH
- 0a32AH :
```

S

Figure S2. (continued)



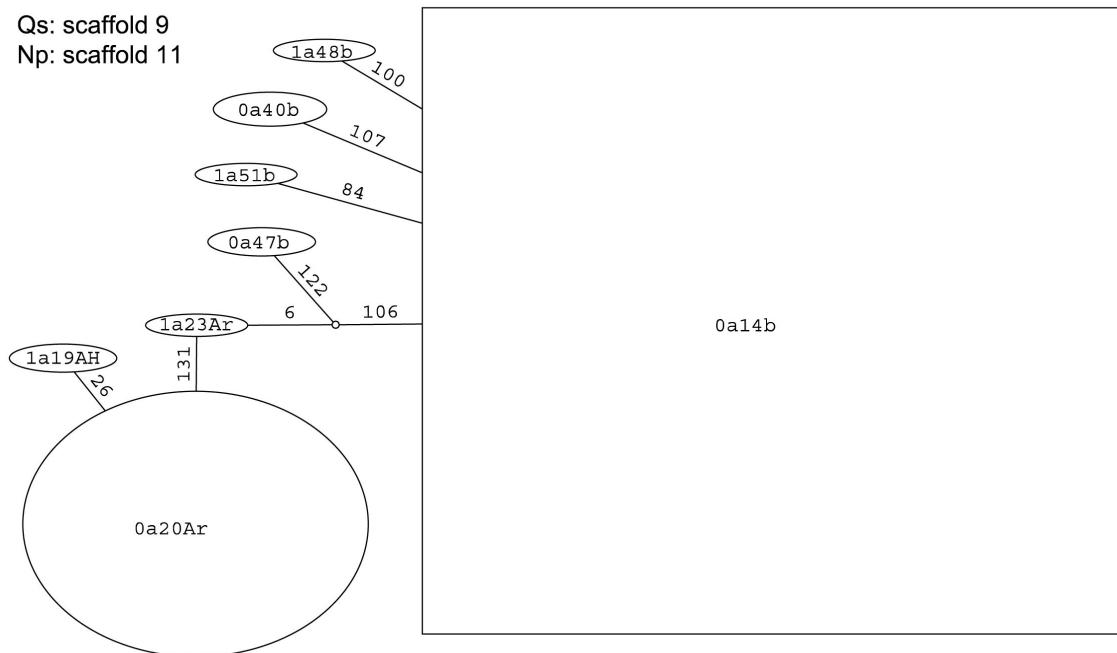
Haplotype list:

```
- 0a3Cb : 1a3Cb 0a15Ab 1a15Ab 1a31Ab 0a31Db 1a31Db 1a32Bb 1a39b 0a40b 1a40b 0a47b
          1a47b 0a52Ab 1a52Ab 0a52Bb 1a52Bb 1a3BH 1a3AH 1a22H 1a27AH
- 0a5Db : 1a5Db 0a8Db 1a8Db 0a9Db 1a9Db 0a9Eb 1a9Eb 0a9Fb 1a9Fb 0a9Hb 1a9Hb
          0a10Ab 1a10Ab 0a10Bb 1a10Bb 1a4AH 1a5AH 1a5BH 1a5CH 1a5EH 1a6AH 1a6BH
          1a6EH 1a8AH 1a8BH 1a8CH 1a9BH 1a9CH 1a30AH 1a30CH
- 0a31Ab : 0a31Cb 1a31Cb 1a37Bb 0a44b 1a44b 1a45Ab 0a45Bb 1a45Bb 0a46b
- 0a32Bb : 0a35b 1a35b 0a36b 1a36b 1a49Bb 1a29BH 1a34H
- 0a37Bb : 0a45Ab 0a48b 1a48b 0a49Bb 0a33H
- 0a39b :
- 1a46b :
- 0a51b : 1a51b
- 0a20Ar : 1a20Ar 0a20Br 1a20Br 0a21Ar 1a21Ar 0a21Br 1a21Br 0a23Br 1a23Br 0a24Ar 1a24Ar
          0a24Br 1a24Br 1a25Ar 0a25Br 1a25Br 0a30Br 1a30Br 0a30Dr 1a30Dr 0a3BH 0a4AH
          0a5AH 0a5BH 0a5CH 0a5EH 0a6AH 0a6BH 0a6EH 0a8AH 0a8BH 0a8CH 0a9BH
          0a9CH 0a9GH 0a3AH 0a19AH 0a22H 0a27AH 0a27BH 0a29BH 0a30AH 0a30CH 1a33H
          0a34H
- 0a25Ar :
- 1a9GH :
- 1a19AH :
- 1a27BH :
```

T

Figure S2. (continued)

Qs: scaffold 9
Np: scaffold 11



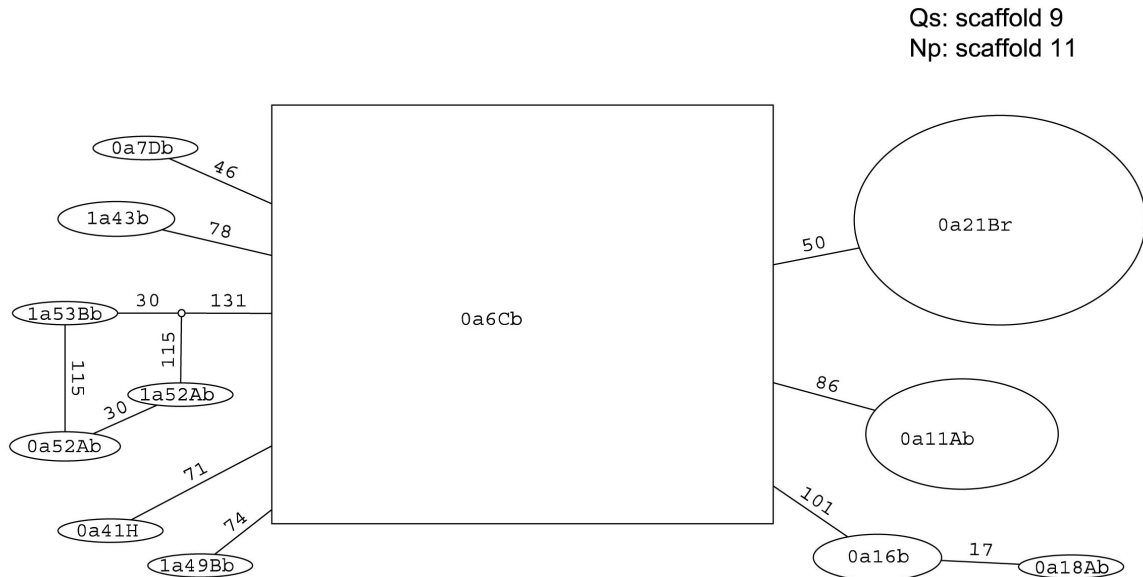
Haplotype list:

```
- 0a14b : 1a14b 0a3Cb 1a3Cb 0a5Db 1a5Db 0a6Cb 1a6Cb 0a6Fb 1a6Fb 0a8Db 1a8Db
          0a9Ab 1a9Ab 0a9Db 1a9Db 0a9Fb 1a9Fb 0a9Hb 1a9Hb 0a10Ab 1a10Ab 0a10Bb
          1a10Bb 0a12b 1a12b 0a13b 1a13b 0a15Ab 1a15Ab 0a15Bb 1a15Bb 0a18Ab 1a18Ab
          0a31Ab 1a31Ab 0a31Bb 1a31Bb 0a31Cb 1a31Cb 0a31Db 1a31Db 0a32Bb 1a32Bb 0a35b
          1a35b 0a36b 1a36b 0a37Ab 1a37Ab 0a37Bb 1a37Bb 0a38Ab 1a38Ab 0a38Bb 1a38Bb
          0a39b 1a39b 0a42Bb 1a42Bb 0a44b 1a44b 0a45Ab 1a45Ab 0a45Bb 1a45Bb 0a48b
          0a49Ab 1a49Ab 0a49Bb 1a49Bb 0a50b 1a50b 0a51b 0a52Ab 1a52Ab 0a52Bb 1a52Bb
          0a53Ab 1a53Ab 0a53Bb 1a53Bb 0a42Ab 1a42Ab 0a3BH 0a5BH 0a5CH 0a6AH 0a6BH
          0a6DH 0a8AH 0a8BH 0a8CH 0a9BH 0a9CH 0a9GH 0a3AH 0a19AH 0a19BH 0a22H
          0a27AH 0a27BH 0a29AH 0a29BH 0a30AH 0a30CH 0a32AH 0a33H

- 0a40b : 1a40b 0a41H
- 0a47b : 1a47b
- 1a48b :
- 1a51b :
- 0a20Ar : 1a20Ar 0a20Br 1a20Br 0a21Ar 1a21Ar 0a21Br 1a21Br 0a23Ar 0a23Br 1a23Br 0a24Ar
          1a24Ar 0a25Ar 1a25Ar 0a25Br 1a25Br 0a30Br 1a30Br 0a30Dr 1a30Dr 1a3BH 1a5BH
          1a5CH 1a6AH 1a6BH 1a6DH 1a8AH 1a8BH 1a8CH 1a9BH 1a9CH 1a9GH 1a3AH
          1a22H 1a27AH 1a27BH 1a29AH 1a29BH 1a30AH 1a30CH 1a32AH 1a33H 1a41H
- 1a23Ar :
- 1a19AH : 1a19BH
```

U

Figure S2. (continued)



Haplotype list:

```

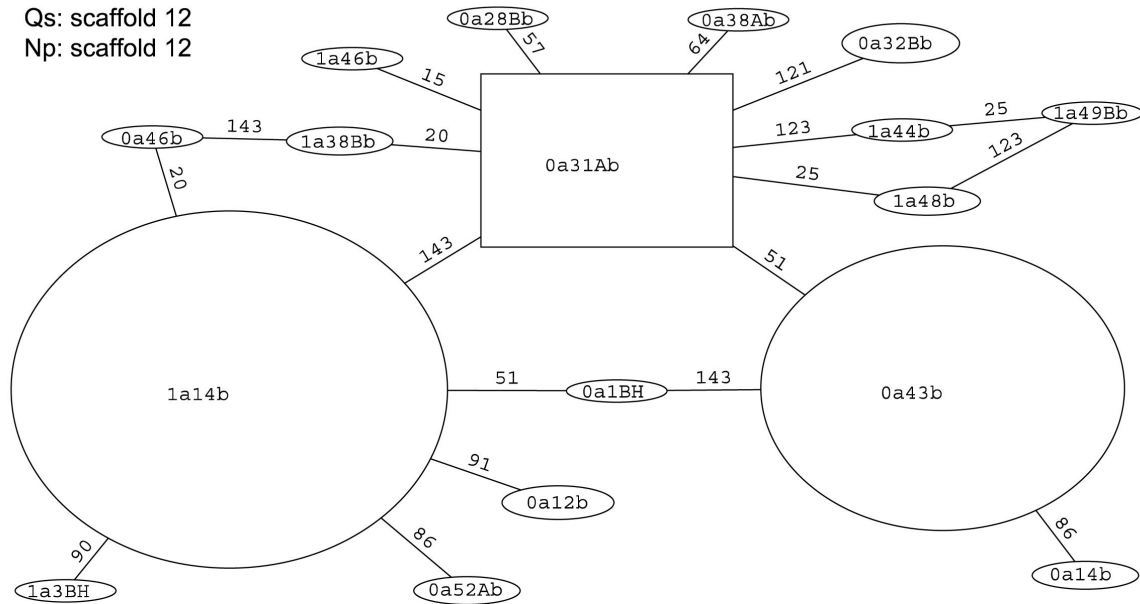
- 0a6Cb : 1a6Cb 0a7Bb 1a7Bb 1a7Db 0a8Db 1a8Db 0a9Ab 1a9Ab 0a9Db 1a9Db 0a9Fb
          1a9Fb 0a9Hb 1a9Hb 0a28Bb 1a28Bb 0a31Ab 1a31Ab 0a31Bb 1a31Bb 0a31Cb 1a31Cb
          0a31Db 1a31Db 0a32Bb 1a32Bb 0a35b 1a35b 0a36b 1a36b 0a37Ab 1a37Ab 0a37Bb
          1a37Bb 0a38Bb 1a38Bb 0a39b 1a39b 0a40b 1a40b 0a43b 0a44b 1a44b 0a46b
          1a46b 0a49Ab 1a49Ab 0a49Bb 0a50b 1a50b 0a51b 1a51b 0a42Ab 1a42Ab 0a5CH
          0a6AH 0a7AH 0a7CH 0a8BH 0a9BH 0a9CH 0a9GH 0a18BH 0a27AH 0a27BH 0a29BH
          0a32AH 0a33H
- 0a7Db :
- 0a11Ab : 1a11Ab 0a11Bb 1a11Bb 0a12b 1a12b 0a13b 1a13b 0a2H 0a4BH 0a5AH 0a5BH
          0a5EH 0a6EH 0a1AH 0a1BH
- 0a16b : 1a16b 0a17b 1a17b 1a18Ab
- 0a18Ab :
- 1a43b : 0a26AH 0a26BH
- 1a49Bb :
- 0a52Ab : 0a53Bb
- 1a52Ab :
- 1a53Bb :
- 0a21Br : 1a21Br 0a25Br 1a25Br 0a30Br 1a30Br 0a30Dr 1a30Dr 1a2H 1a4BH 1a5AH 1a5BH
          1a5CH 1a5EH 1a6AH 1a6EH 1a7AH 1a7CH 1a8BH 1a9BH 1a9CH 1a9GH 1a18BH
          1a26AH 1a26BH 1a27AH 1a27BH 1a29BH 1a32AH 1a33H 1a41H 1a1AH 1a1BH
- 0a41H :

```

V

Figure S2. (continued)

Qs: scaffold 12
Np: scaffold 12



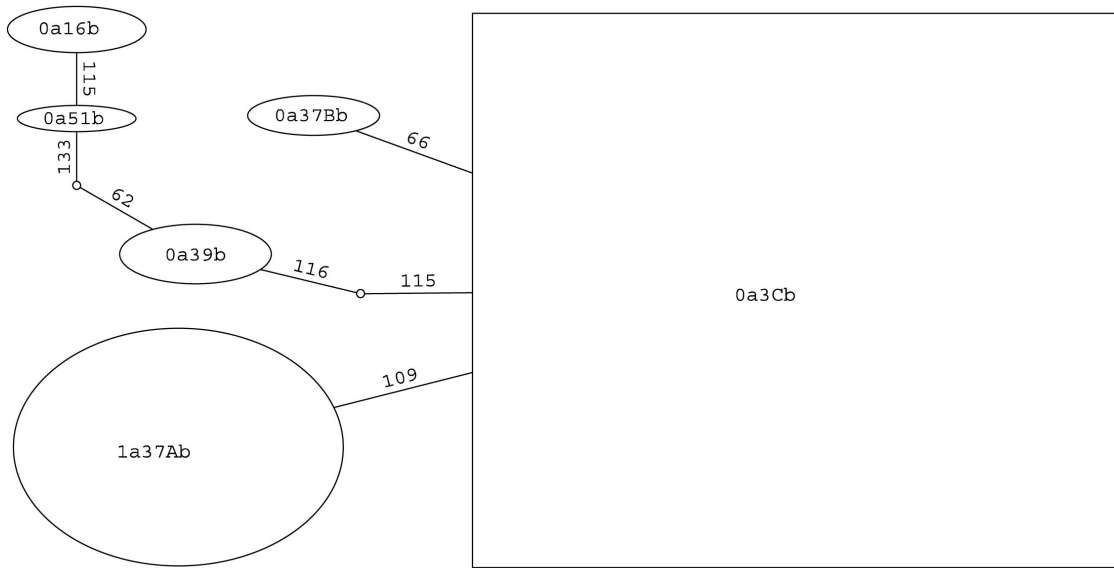
Haplotype list:

- 0a14b : 0a15Ab
- 1a14b : 0a5Db 1a5Db 0a6Cb 1a6Cb 0a6Fb 1a6Fb 0a7Bb 1a7Bb 0a8Db 1a8Db 0a9Ab 1a9Ab 0a9Db 1a9Db 0a9Eb 1a9Eb 0a9Fb 1a9Fb 0a9Hb 1a9Hb 0a10Ab 1a10Ab 0a11Ab 1a11Ab 0a11Bb 1a11Bb 0a13b 1a13b 1a15Ab 0a18Ab 1a18Ab 1a28Bb 0a37Ab 1a37Ab 0a37Bb 1a37Bb 0a40b 0a44b 0a49Ab 1a4AH 1a4BH 1a5AH 1a5BH 1a5CH 1a5EH 1a6AH 1a6BH 1a6DH 1a8AH 1a8BH 1a8CH 1a9BH 1a9CH 1a9GH 1a19AH 1a22H 1a27BH 1a28AH 1a1AH 1a1BH
- 0a12b : 1a12b 1a30CH
- 0a28Bb :
- 0a31Ab : 1a31Ab 0a31Bb 1a31Bb 0a31Cb 1a31Cb 0a31Db 1a31Db 1a32Bb 1a36b 1a38Ab 0a38Bb 0a42Bb 1a42Bb 1a43b 1a45Bb 0a48b 1a49Ab 0a50b 1a50b 0a53Bb 1a53Bb 1a26AH 1a26BH 1a27AH 1a29AH 1a29BH 1a41H
- 0a32Bb : 0a36b 1a32AH 1a34H
- 0a38Ab :
- 1a38Bb : 1a40b
- 0a43b : 0a45Bb 0a21Br 1a21Br 0a23Ar 1a23Ar 0a23Br 1a23Br 0a24Ar 1a24Ar 0a25Ar 1a25Ar 0a25Br 1a25Br 0a30Br 1a30Br 0a30Dr 1a30Dr 0a3BH 0a4AH 0a4BH 0a5AH 0a5BH 0a5CH 0a5EH 0a6AH 0a6BH 0a6DH 0a8AH 0a8BH 0a8CH 0a9BH 0a9CH 0a9GH 0a19AH 0a22H 0a26AH 0a26BH 0a27AH 0a27BH 0a28AH 0a29AH 0a29BH 0a30CH 0a32AH 0a34H 0a41H 0a1AH
- 1a44b :
- 0a46b :
- 1a46b :
- 1a48b : 0a49Bb
- 1a49Bb :
- 0a52Ab : 1a52Ab
- 1a3BH :
- 0a1BH :

W

Figure S2. (continued)

Qs: scaffold 12
Np: scaffold 12



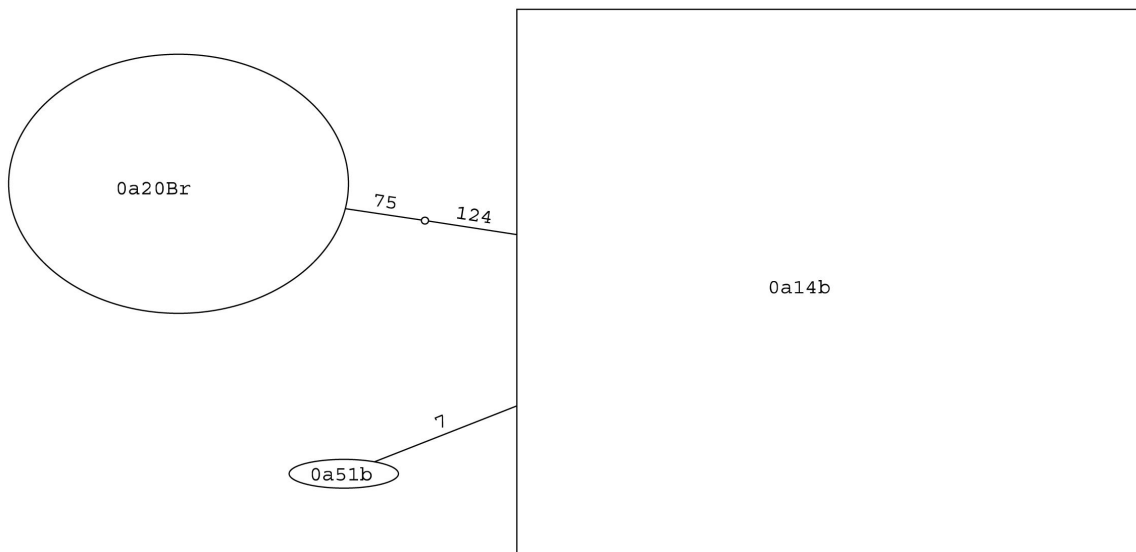
Haplotype list:

```
- 0a3Cb : 1a3Cb 0a5Db 1a5Db 0a6Cb 1a6Cb 0a6Fb 1a6Fb 0a7Bb 1a7Bb 0a7Db
          1a7Db 0a9Ab 1a9Ab 0a9Db 1a9Db 0a9Eb 1a9Eb 0a9Fb 1a9Fb 0a9Hb
          1a9Hb 0a11Ab 1a11Ab 0a12b 1a12b 0a28Bb 1a28Bb 0a32Bb 1a32Bb 0a35b
          1a35b 0a36b 1a36b 0a37Ab 1a37Bb 0a38Bb 1a38Bb 1a39b 0a42Bb 1a42Bb
          0a43b 1a43b 0a44b 1a44b 0a45Ab 1a45Ab 0a48b 1a48b 0a49Ab 1a49Ab
          0a50b 1a50b 0a53Ab 1a53Ab 0a53Bb 1a53Bb 0a2H 0a4BH 0a5AH 0a5BH
          0a5CH 0a5EH 0a6AH 0a6DH 0a7AH 0a7CH 0a8BH 0a8CH 0a9GH 0a22H
          0a26AH 0a26BH 0a29BH 0a30AH 0a30CH 0a32AH 0a34H 0a41H 0a1AH 0a1BH
- 0a16b : 1a16b 0a17b 1a17b
- 1a37Ab : 0a21Br 1a21Br 0a24Ar 1a24Ar 0a30Br 1a30Br 0a30Dr 1a30Dr 1a2H 1a4BH
          1a5AH 1a5BH 1a5CH 1a5EH 1a6AH 1a6DH 1a7AH 1a7CH 1a8BH 1a8CH
          1a9GH 1a22H 1a26AH 1a26BH 1a29BH 1a30AH 1a30CH 1a32AH 1a34H 1a41H
          1a1AH 1a1BH
- 0a37Bb : 0a52Ab 1a52Ab
- 0a39b : 0a40b 1a40b 0a49Bb 1a49Bb 1a51b
- 0a51b :
```

X

Figure S2. (continued)

Qs: scaffold 13
Np: scaffold 13

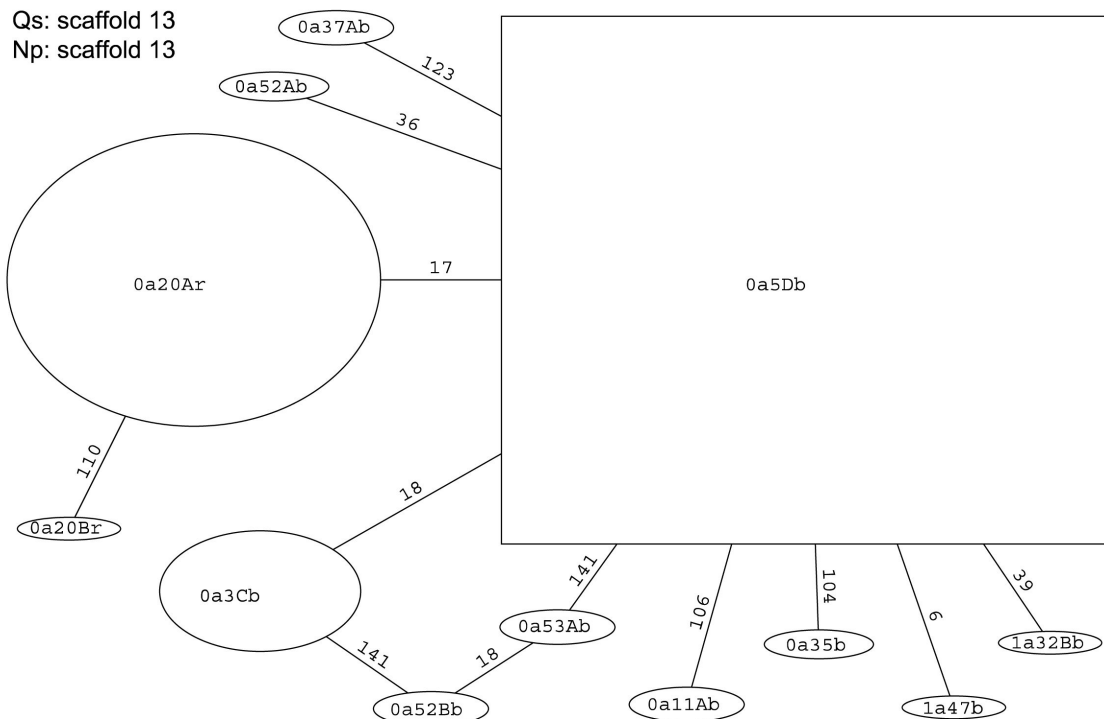


Haplotype list:

```
- 0a14b : 1a14b 0a3Cb 1a3Cb 0a5Db 1a5Db 0a6Cb 1a6Cb 0a6Fb 1a6Fb 0a8Db
          1a8Db 0a9Ab 1a9Ab 0a9Db 1a9Db 0a9Eb 1a9Eb 0a9Fb 1a9Fb 0a9Hb
          1a9Hb 0a10Bb 1a10Bb 0a13b 1a13b 0a15Ab 1a15Ab 0a18Ab 1a18Ab 0a31Ab
          1a31Ab 0a31Bb 1a31Bb 0a31Cb 1a31Cb 0a31Db 1a31Db 0a35b 1a35b 0a37Ab
          1a37Ab 0a37Bb 1a37Bb 0a40b 1a40b 0a45Ab 1a45Ab 0a45Bb 1a45Bb 0a46b
          1a46b 0a47b 1a47b 0a48b 1a48b 0a49Ab 1a49Ab 0a49Bb 1a49Bb 1a51b
          0a52Ab 1a52Ab 1a52Bb 0a53Ab 1a53Ab 0a53Bb 1a53Bb 1a3BH 1a5AH 1a5BH
          1a5CH 1a5EH 1a6AH 1a6BH 1a6DH 1a6EH 1a8AH 1a8BH 1a8CH 1a9BH
          1a9CH 1a9GH 1a3AH 1a26AH 1a27AH 1a27BH 1a29BH 1a30CH 1a32AH 1a34H
          1a41H
- 0a51b : 0a52Bb
- 0a20Br : 1a20Br 0a21Ar 1a21Ar 0a23Ar 1a23Ar 0a23Br 1a23Br 0a24Br 1a24Br 0a25Ar
          1a25Ar 0a25Br 1a25Br 0a30Br 1a30Br 0a30Dr 1a30Dr 0a3BH 0a5AH 0a5BH
          0a5CH 0a5EH 0a6AH 0a6BH 0a6DH 0a6EH 0a8AH 0a8BH 0a8CH 0a9BH
          0a9CH 0a9GH 0a3AH 0a26AH 0a27AH 0a27BH 0a29BH 0a30CH 0a32AH 0a34H
          0a41H
```

Y

Figure S2. (continued)



Haplotype list:

```

- 0a3Cb : 1a3Cb 0a31Bb 1a31Bb 0a31Cb 0a31Db 1a31Db 0a32Bb 0a38Bb 0a40b 1a40b 0a42Bb
          0a45Bb 0a48b 1a48b 0a26AH 0a27AH 0a27BH
- 0a5Db : 1a5Db 0a6Cb 1a6Cb 0a6Fb 1a6Fb 0a7Bb 1a7Bb 0a7Db 1a7Db 0a8Db 1a8Db
          0a9Ab 1a9Ab 0a9Db 1a9Db 0a9Eb 1a9Eb 0a9Fb 1a9Fb 0a9Hb 1a9Hb 0a10Bb
          1a10Bb 1a11Ab 0a16b 1a16b 0a17b 1a17b 0a18Ab 1a18Ab 0a28Bb 1a28Bb 1a31Cb
          1a35b 1a36b 1a37Bb 0a38Ab 1a38Ab 1a38Bb 0a39b 1a39b 1a42Bb 0a44b 1a44b
          0a45Ab 1a45Ab 1a45Bb 0a46b 1a46b 0a47b 0a49Ab 1a49Ab 0a49Bb 1a49Bb 0a51b
          1a51b 0a42Ab 1a42Ab 0a2H 0a4BH 0a5BH 0a5CH 0a5EH 0a6AH 0a6BH 0a6DH
          0a6EH 0a7AH 0a7CH 0a8AH 0a8BH 0a8CH 0a9BH 0a9CH 0a9GH 0a3AH 0a19BH
          0a22H 0a26BH 0a28AH 0a29BH 0a30AH 0a30CH 0a32AH 0a34H 0a41H 0a1AH 0a1BH
- 0a11Ab : 0a11Bb 1a11Bb
- 1a32Bb :
- 0a35b : 0a36b
- 0a37Ab : 1a37Ab 0a37Bb
- 1a47b :
- 0a52Ab : 1a52Ab
- 0a52Bb : 1a52Bb 0a53Bb
- 0a53Ab : 1a53Ab 1a53Bb
- 0a20Ar : 1a20Ar 1a20Br 0a21Br 1a21Br 0a23Br 1a23Br 0a24Ar 1a24Ar 0a25Ar 1a25Ar 0a30Br
          1a30Br 0a30Dr 1a30Dr 1a2H 1a4BH 1a5BH 1a5CH 1a5EH 1a6AH 1a6BH 1a6DH
          1a6EH 1a7AH 1a7CH 1a8AH 1a8BH 1a8CH 1a9BH 1a9CH 1a9GH 1a3AH 1a19BH
          1a22H 1a26AH 1a26BH 1a27AH 1a27BH 1a28AH 1a29BH 1a30AH 1a30CH 1a32AH 1a34H
          1a41H 1a1AH 1a1BH
- 0a20Br :

```

Z

Figure S2. (continued)

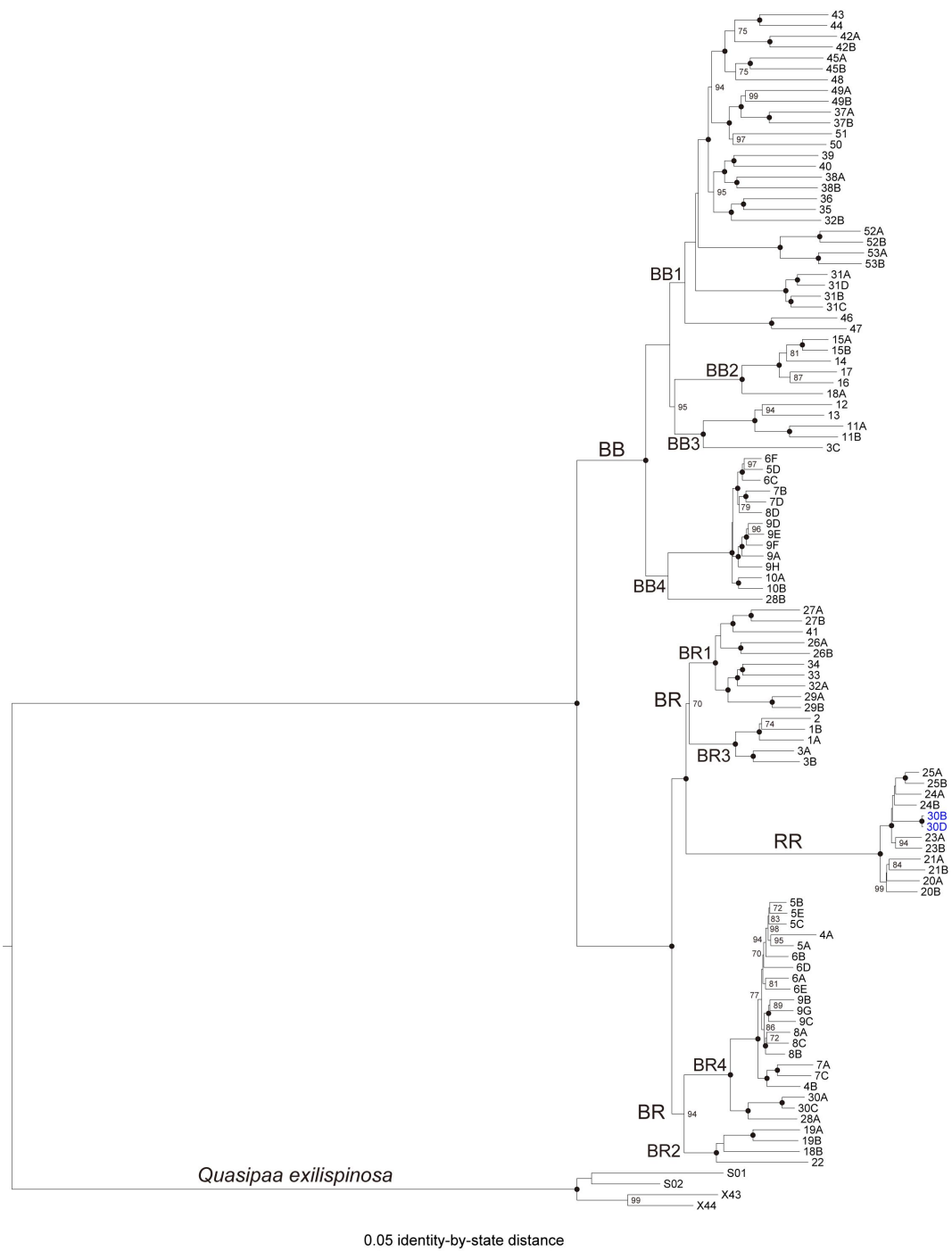


Figure S3.

Neighbor-joining tree based on the SNP identity-by-state distance matrix for individuals.

Numbers at nodes are bootstrap support values ≥ 70 , and circles indicate support values of 100. BB: *Quasipaa boulengeri*; RR: *Q. robertingeri*; BR: putative hybridogenetic hybrids between *Q. boulengeri* and *Q. robertingeri*.

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Additional files

Table S1. Details and proportion of heterozygous loci (H_o) of samples. The main dataset contained a subset of 20,912 SNPs each corresponding to a locus with unique BLAST hits on homologous scaffolds of a *Quasipaa spinosa* genome and a *Nanorana parkeri* genome. BR: putative *Q. boulengeri* × *Q. robertingeri* hybridogen; IRT: karyotype with an interchromosomal reciprocal translocation. [↗](#)

Table S2. Primers used in PCR and sequencing of fragments of *Quasipaa boulengeri* mitochondrial genome. [↗](#)

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Reviewer #1 (Public review):

Summary:

(1) Introduction Hybridogenesis involves one genome being clonally transmitted while the other is replaced by backcrossing. It results in high heterozygosity and balanced ancestry proportions in hybrids. Distinguishing it from other hybrid systems requires a combination of nuclear, mitochondrial, and population-genetic evidence. Hybridogenesis has been identified in only a few taxa (e.g., some fish, frogs, and stick insects), but no new cases have been reported in over a decade. Advancements in high-throughput sequencing now allow for the detection of high individual heterozygosity, which can indicate hybridization, but it is difficult to distinguish hybridogenesis from other similar asexual systems based solely on genome-wide data. To differentiate these systems, researchers look at several key indicators: Presence of pure-species offspring from hybrids (possible only in hybridogenesis); sex ratio (male presence in hybridogenetic systems); nuclear and mitochondrial haplotype sharing with co-distributed parental species; geographic distribution patterns, especially the lack of both parental species in hybrid populations.

(2) What the authors were trying to achieve The paper studies *Quasipaa* Frogs. *Q. robertingeri* (narrowly endemic) and *Q. boulengeri* (widespread), which are morphologically similar and found sympatrically in parts of China. Preliminary RAD-seq data revealed bimodal heterozygosity in *Q. boulengeri* samples. Some individuals had extremely high heterozygosity, consistent across loci and suggestive of F1 hybrids. These high-heterozygosity individuals had one haplotype from each species. The study investigates the high heterozygosity observed in *Quasipaa* frogs, particularly in individuals morphologically resembling *Q. boulengeri* but genetically appearing to be F1 hybrids with *Q. robertingeri*. The

goal is to determine whether these patterns are consistent with hybridogenesis, rather than other atypical reproductive modes. The authors also suggest the hypothesis that hybridogenesis could enable range expansion of an endemic species through hybridization with a widespread relative.

(3) Methods A total of 107 individuals from 53 localities were collected for the study. This sample included 58 sexed adults—27 males and 31 females—as well as a majority of tadpoles. Of these individuals, 31 had previously determined karyotypes. DNA was extracted and sequenced. Individual heterozygosity and ancestry were estimated using bioinformatics tools. F1 hybrids were compared to one of the parental species to examine patterns of fixed heterozygous loci. Mitochondrial DNA was also extracted from sequencing data, and phylogenetic trees were constructed

(4) Results Two groups of individuals were detected based on heterozygosity: one group exhibited high heterozygosity and consisted of F1 hybrids, while the other group showed low heterozygosity, representing pure-species types. The F1 hybrids demonstrated approximately equal ancestry from *Q. robertingeri* and *Q. boulengeri*, consistently maintaining a high proportion of heterozygous loci at around 16.7%. In contrast, pure individuals had much lower heterozygosity, approximately 2.9%. F1 hybrids were found across 21 different sites, including both male and female individuals. The presence of numerous fixed heterozygous loci in F1 hybrids confirmed their hybrid origin, and these loci were absent in pure *Q. boulengeri* samples. F1 individuals typically carried one haplotype from each parental species. There was minimal haplotype sharing between the two pure species, but extensive sharing was observed between F1 hybrids and co-occurring pure-species individuals. In fact, F1 types shared haplotypes with local *Q. boulengeri* in over 90% of cases, which supports the occurrence of local backcrossing and parental contribution. In terms of mitochondrial DNA, F1 hybrids possessed mitochondrial haplotypes that clustered with *Q. boulengeri* and often shared these haplotypes directly. Genetic structure and phylogenetic analyses, revealed three distinct genetic clusters corresponding to F1 hybrids, *Q. boulengeri*, and *Q. robertingeri*. The F1 hybrids positioned themselves intermediate between the two pure species. Neighbor-joining trees and TreeMix analyses confirmed a strong separation between pure-species types, with F1 hybrids clustering alongside local *Q. boulengeri* subpopulations, indicating local formation of hybrids.

(5) Discussion In summary, the study reveals hybridogenesis (a reproductive system where hybrids clonally transmit one parental genome) in *Quasipaa boulengeri* and *Q. robertingeri*. Hybrids show high genetic heterozygosity and coexist with parental species, ruling out other reproductive modes like parthenogenesis or kleptogenesis. Evidence suggests hybridogenesis enables *Q. robertingeri* genomes to appear far outside their normal range, possibly aiding range expansion. Chromosomal abnormalities are linked to hybrid hybrids, supporting clonal genome transmission. The genetic divergence between parental species fits patterns seen in other hybridogenetic systems, highlighting a unique, understudied case in East Asia.

Strengths:

Overall, the authors carefully interpret their genetic data to support hybridogenesis as the reproductive mode in this system and propose that this mechanism may aid range expansion. They also appropriately acknowledge the need for further cytogenetic and ecological studies, demonstrating scientific caution. In summary, the discussion reasonably follows from the results, offering cautious interpretation where necessary.

Weaknesses:

Direct reproductive or cytological evidence is still lacking. While alternative reproductive modes are discussed and mostly ruled out logically, some require further empirical testing.

The authors maintain a cautious interpretation, appropriately suggesting further research. Some outstanding questions remain.

(1) The elevated heterozygosity and presence of fixed heterozygous loci in hybrids compared to parental species strongly indicate hybridogenesis. However, alternative explanations such as repeated F1 hybridization or some form of balanced polymorphism, while less likely, are not fully excluded.

(2) The coexistence of hybrids and parental species, along with high nuclear and mitochondrial haplotype sharing between hybrids and *Q. boulengeri*, argues against reproductive modes like parthenogenesis, gynogenesis, or kleptogenesis. However, the assumption that hybrid sterility or multiple local hybrid origins are unlikely could be challenged if undetected local variation or cryptic reproductive strategies exist.

(3) The presence of *Q. robertingeri* nuclear genomes far outside their known geographic range, genetically linked to nearby populations, fits a hybridogenetic-mediated dispersal model. Although the authors dismiss human-mediated or accidental transport as explanations, these scenarios are not necessarily unlikely.

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Reviewer #2 (Public review):

This study describes F1 hybrid frog lineages that use an "unusual" form of reproduction, perhaps hybridogenesis. Identifying such species is important for understanding the biodiversity of reproduction in animals, and animals that do not reproduce via "canonical" sex can be useful model systems in ecology and evolution. The conclusion of the study are based on reduced representation sequencing (RAD-seq with a de-novo assembly of loci) of 107 wild-caught individuals from 53 localities (plus 4 outgroup individuals), including 27 males, 31 females, and 49 juveniles of unknown sex. Conclusive inferences of unusual forms of reproduction typically require breeding studies and parent-offspring genotype comparisons but such information is not available (and perhaps impossible to generate) for the focal frog lineages.

(1) Conclusion 1: there are two pure species and F1 hybrids

The authors infer that there are two lineages RR and BB (corresponding to two named species), and F1 interspecific hybrids RB. This inference is based on the results presented in Figure 1 (PCA, admixture, and heterozygosity analyses) as well as analyses of fixed SNP differences between R and B. I think that this conclusion is well supported; my only comment on this part is that it would be useful to have the admixture plots & cross-validation for the 107 samples with other k values (not only $k=2$) as a supplemental figure. The plots in the supplemental file S1 are for the subset of 55 inds inferred to be BB only.

(2) Conclusion 2: F1 hybrids most likely reproduce via hybridogenesis

This conclusion is based on the sex ratio of hybrids and haplotype sharing between species and lineages at different, ~150 bp long loci. Parthenogenesis (including sperm-dependent parthenogenesis) is unlikely to generate males, yet sexed F1 hybrid individuals include 18 females and 10 males which prompts the exclusion of parthenogenesis in the present paper. Specific haplotype-sharing patterns are also discussed in the study and used as further support, but these arguments (and the related main and supplementary figures) are difficult to read/interpret. To clarify the arguments related to haplotype sharing and haplotype diversities, I suggest that the authors phase the R and B haplotypes from all their hybrids by using their pure (RR and BB individuals) as references. The concatenated lineage-specific haplotypes can then be used to reconstruct a single phylogenetic tree for all loci (easier to

visualize and interpret that the separate haplotype networks for the loci). The authors can then draw cartoon phylogenies for what would be the expected pattern for haplotype clustering and diversity for different reproductive modes, and discuss their observed phylogenies in this regard. Similarly, the migration weights (represented in Figure 4) can then also be computed for separate haplotypes in the hybrids.

However, independently of the outcome of the phasing, it is important to note that there is no a priori reason why all F1 hybrid individuals would reproduce via the same reproductive mode. Notably, work by Barbara Mantovani and Valerio Scali on stick insects has shown that different F1 hybrid lineages involving the same parental species reproduce via hybridogenesis or parthenogenesis. I don't see how the presented data can allow excluding that some F1 hybrid frogs are parthenogenetic while others are hybridogenetic for example.

(3) Conclusion 3: Crosses between hybridogenetic RB males and hybridogenetic RB females gave rise to a new population of RR individuals outside of the RR species range (this new population would correspond to location 30 from Figure 1).

It is not entirely clear to me which data this conclusion is based on, I believe it is the combination of known species ranges for the species R (location 30 being outside of this) and the relatively low heterozygosity of RR individuals at location 30.

However, as the authors point out, the study focuses on an understudied geographic range. Isolated or rare populations of the R species may easily have been overlooked in the past, especially since the R and B species are morphologically difficult to distinguish. Furthermore, an isolated, perhaps vestigial population may also likely be inbred/feature low diversity. It seems most appropriate to discuss different (equally likely) scenarios for the RR population at location 30 rather than implying a hybridogenetic origin of RR individuals. I would also choose a title that does not directly imply this scenario but reflects the solid (not speculative) findings of the study.

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Reviewer #3 (Public review):

Summary:

This work reports a new case of hybridogenetic reproduction in the frog genus *Quasipaa*. Only one other example of this peculiar reproductive mode is known in amphibians, and fewer than a dozen across the tree of life. Interestingly, a population of one of the parental species (*Q. robertingeri*) was found away from the core of its distribution, within the distribution of the hybridogens. This range expansion might have been mediated by hybridogenesis, whereby two copies of the same parental genome came together again after many generations of hybridogenesis.

Strengths:

Evidence for hybridogenesis is solid. The state of the art would be to genotype parents and offspring, but other known alternative scenarios have been considered carefully and can be ruled out convincingly. In addition, the authors are very careful in their phrasing and made sure to never overinterpret their data.

The explicit predictions under different reproductive modes (and Table 1) are a useful resource for future studies and could inspire new findings of unusual reproductive modes in other taxa.

The sampling is very impressive, with over 50 populations sampled across a very large area.

The comparison of p-distances between pairs of species involved in hybridogenesis is interesting.

Weaknesses:

The current phylogenetic reconstruction with the F1s does not enable to infer the number of origins of hybridogenesis, nor whether the population of *Q. robertingeri* that was found far from the core of the species' distribution indeed derives from hybridogenesis. This is because some of the signal is driven by the *Q. boulengeri* haplome, which is replaced every generation and therefore does not reflect the evolutionary history of the lineage.

All known reproductive modes except hybridogenesis can be excluded, but without genotyping parents and offspring, it is impossible to rule out another, yet undescribed reproductive mode.

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